

Team 22257 — Meeting Task Cards

Herobotix - Heritage Academy Gateway · 2026-2027 BIOBUZZ · REV Mecanum Drivetrain Kit v2 (REV-45-2470)

How this works

Every meeting starts with **10 minutes** of orientation together and ends with **10 minutes** of cleanup together. In between you pick a card and work it. **Most cards are built to fill 100 minutes** (a few shorter ones say so on the card) — that is a whole Tuesday (4:00–6:00). Friday (12:00–3:00) gives you 160 minutes, so you get your card plus another hour: finish it properly, do the “If you finish early” box, or start the next card in your chain. Work the checklist top to bottom and get the “Done when” boxes checked before cleanup. If a card says you need another card first, do that one first.

ENGINEERING & PLANNING 13

EP-1 Team Identity: Name, Look &

Motto Start here

NO HARDWARE

EP-2 Read the Game, Build the Wall

Poster Start here

NO HARDWARE

EP-3 Robot Requirements List Some

experience

NO HARDWARE

needs EP-2

EP-4 Design Sketches & Pick a

Concept Some experience

NO HARDWARE

needs EP-3

EP-5 Set Up the Engineering Notebook Start

here

NO HARDWARE

EP-6 Tape Out the Practice Field Start here

NO HARDWARE

EP-7 Shop Safety & Tool Check-Off Start here

NO HARDWARE

EP-8 Scouting & Alliance Strategy Some

experience

NO HARDWARE

needs EP-2

EP-9 Drive Team: Roles, Calls &

Practice Some experience

ROBOT + FIELD

needs PR-19

EP-10 Pit Setup & Event Day Run Sheet Start

here

NO HARDWARE

EP-11 Judging: Portfolio & the Five-Minute

Interview Challenge

NO HARDWARE

needs EP-5

EP-12 Post-Event Debrief: What Mk-2 Must

Fix Some experience

NO HARDWARE

needs EP-8

EP-13 Build the BIOBUZZ Field

Elements Some experience

NO HARDWARE

needs EP-7

CAD & 3D PRINTING 9

CD-1 Onshape Setup & Your First Part Start

here

NO HARDWARE

CD-2 Measure & Model a Real Robot

Part Some experience

NO HARDWARE

needs CD-1

CD-3 Design a Pollen Intake in

CAD Challenge

NO HARDWARE

needs CD-2, EP-4

CD-4 Design a Pollen Outtake in

CAD Challenge

NO HARDWARE

needs CD-2, EP-2

CD-5 Build the Robot in CAD: Assembly &

Fit Check Challenge

NO HARDWARE

needs CD-2

CD-6 From CAD to Printed Part Some

experience

NO HARDWARE

needs CD-2

CD-7 Design a Printed Part That Solves a

Real Problem Challenge

NO HARDWARE

needs CD-6

CD-8 Design the Odometry Mounting

Adapter Challenge

NO HARDWARE

needs CD-2, BA-3

CD-9 Fit the Starter Bot to Our Chassis in

CAD Challenge

NO HARDWARE

needs CD-5

PROGRAMMING 21

PR-1 Get Connected: Control Hub & Driver Hub [Start here](#)

BOARD OR ROBOT

PR-2 Build the Robot Configuration File [Start here](#)

BOARD OR ROBOT

needs PR-1

PR-3 First TeleOp: Make It Drive (Mecanum) [Start here](#)

ROBOT

needs PR-2

PR-4 Telemetry: See What the Robot Is Thinking [Some experience](#)

BOARD OR ROBOT

needs PR-3

PR-5 Driver Tuning: Slow Mode & Smooth Control [Some experience](#)

ROBOT + FIELD

needs PR-3, EP-6

PR-6 First Autonomous: Time-Based Movement [Some experience](#)

ROBOT + FIELD

needs PR-3, EP-6

PR-7 Encoder Autonomous: Drive an Exact Distance [Challenge](#)

ROBOT + FIELD

needs PR-6, PR-4

PR-8 Intake Control: Map a Mechanism to a Button [Some experience](#)

ROBOT

needs PR-2, BA-5

PR-9 Practice Without the Robot: FTC SIM [Start here](#)

NO HARDWARE

PR-10 Touch Sensor: The First Decision [Start here](#)

PROGRAMMING BOARD

needs PR-2

PR-11 Color Sensor: Teach It to See Pollen [Some experience](#)

PROGRAMMING BOARD

needs PR-4

PROGRAMMING CONT.

PR-12 Potentiometer & Servo: Analog In, Position Out [Some experience](#)

PROGRAMMING BOARD

needs PR-4

PR-13 Bench to Robot: Move Your Code Over [Challenge](#)

ROBOT

needs PR-3, PR-10

PR-14 Field-Centric Drive with the IMU [Challenge](#)

ROBOT + FIELD

needs PR-3, PR-4

PR-15 Read the Pinpoint: Configure & Prove It [Some experience](#)

ROBOT

needs BA-10, PR-4

PR-16 Pedro Pathing: Install & Tune the Localizer [Challenge](#)

ROBOT + FIELD

needs PR-15, PR-3

PR-17 Pedro Pathing: Tune the Follower [Challenge](#)

ROBOT + FIELD

needs PR-16, EP-6

PR-18 Starter Bot Configuration: Six Motors, Two Hubs [Some experience](#)

ROBOT

needs BA-17

PR-19 Run the Starter Bot TeleOp on Mecanum [Challenge](#)

ROBOT + FIELD

needs PR-18, PR-3

PR-20 Flywheel Tuning: Make the Shot Repeatable [Challenge](#)

ROBOT + FIELD

needs PR-19, PR-4

PR-21 Starter Bot Autonomous: Score the Pre-Loads [Challenge](#)

ROBOT + FIELD

needs PR-19

BUILDING & ASSEMBLY 18

BA-1 Inventory the Kit & Set Up the

Shop Start here

NO HARDWARE

BA-2 Mecanum Chassis: Build the

Frame Start here

ROBOT

needs BA-1, EP-7

BA-3 Mecanum Drive: Gearboxes, Motors &

Wheels Some experience

ROBOT

needs BA-2

BA-4 Electronics: Mount & Wire the Control

Hub Some experience

ROBOT

needs BA-3, EP-7

BA-5 Design & Build an Intake Challenge

ROBOT

needs BA-3, EP-4

BA-6 Battery Management Station Start here

NO HARDWARE

BA-7 Scoring Mechanism Prototype

(cardboard first) Challenge

ROBOT

needs BA-5, EP-4

BA-8 Robot Cart & Pit Kit Start here

ROBOT

BA-9 Assemble a Programming Board Start here

PROGRAMMING BOARD

needs EP-7

BUILDING & ASSEMBLY CONT.

BA-10 Install the Pinpoint Odometry

Pods Some experience

ROBOT

needs BA-4, CD-8

BA-11 Starter Bot: Parts Check & Cut

List Some experience

NO HARDWARE

needs BA-1, EP-7

BA-12 Starter Bot: Launcher Floor Some

experience

NO HARDWARE

needs BA-11

BA-13 Starter Bot: Intake Some experience

NO HARDWARE

needs BA-11

BA-14 Starter Bot: Launcher Flywheel &

Feeder Challenge

NO HARDWARE

needs BA-11

BA-15 Starter Bot: Pollen Pusher Start here

NO HARDWARE

needs BA-11

BA-16 Mount the Starter Bot on Our

Mecanum Chassis Challenge

ROBOT

needs BA-12, BA-13, BA-14, BA-15, CD-9

BA-17 Add the Expansion Hub: Power,

Comms & Six Motors Challenge

ROBOT

needs BA-16, BA-4

BA-18 Inspection Dry Run: Make It

Legal Some experience

ROBOT

needs BA-17

CHOOSING CARDS — THE RULES THAT KEEP THIS WORKING

1. **Read “Before you start” first.** If a card names another card, that one has to be finished by somebody first. Most cards can be started cold.
2. **Check the hardware tag.** It decides what can run at the same time. Only the robot cards compete for the robot; the programming board and the “no hardware” cards run alongside them.
3. **Every merch and t-shirt card except TS-3 waits on EP-1.** There is no team name until meeting 1 is done, so there is nothing to put on a shirt.
4. **Match the crew size on the card.** A card marked 2–3 does not absorb six people; it absorbs three and bores the other three. Run a second card instead.
5. **Rotate.** Nobody spends the whole season on one category. The point of a menu is that everyone tries things they would not have picked.
6. **A card is not done because time ran out.** It is done when the “Done when” boxes are true. If it did not finish, note where it stopped and hand the card to the next crew.
7. **Anything leaving the team needs coach approval** — emails to businesses, social posts, orders, anything with a student’s name or face on it.

MERCH & GIVE-AWAY 6

MG-1 Give-Away Plan & Budget Start here

NO HARDWARE

needs EP-1

MG-2 Design the Give-Away Art Some experience

NO HARDWARE

needs EP-1, MG-1

MG-3 Produce the Give-Aways (assembly line) Start here

NO HARDWARE

needs MG-2

MG-4 Team Card / Business Card Some experience

NO HARDWARE

needs EP-1

MG-5 3D-Printed Give-Away Challenge

NO HARDWARE

needs MG-1

MG-6 Design the Award Stickers Some experience

NO HARDWARE

needs EP-1

TEAM T-SHIRTS 5

TS-1 Shirt Design Concepts Start here

NO HARDWARE

needs EP-1

TS-2 Vote & Make the Art Print-Ready Some experience

NO HARDWARE

needs TS-1

TS-3 Sizing Survey & Order Sheet Start here

NO HARDWARE

TS-4 Price the Shirts: Blanks +

Transfers Some experience

NO HARDWARE

needs TS-2, TS-3

TS-5 Press the Team Shirts Some experience

NO HARDWARE

needs TS-4

MARKETING & FUNDRAISING 7

MF-1 Build the Sponsor Prospect List Start here

NO HARDWARE

MF-2 Sponsorship One-Pager Some experience

NO HARDWARE

needs EP-1, MF-1

MF-3 Write the Outreach Emails & Letters Some experience

NO HARDWARE

needs MF-2

MF-4 Social Media Plan & First Posts Start here

NO HARDWARE

needs EP-1

MF-5 Team Page Content for the Herobotix Website Some experience

NO HARDWARE

needs EP-1

MF-6 Plan a Demo / Outreach Event Some experience

ROBOT

needs PR-3

MF-7 Grants & Team Funding

Applications Challenge

NO HARDWARE

needs MF-2

RESOURCE SHELF

Skill Builders online training (FIRST Dashboard login)

<https://training.firstinspires.org/learn/course/ftc-skill-builders-26-27>

REV Mecanum Drivetrain V2 build guide - the student robot

<https://docs.revrobotics.com/duo-build/mecanum-drivetrain-v2>

REV UltraPlanetary gearbox docs

<https://docs.revrobotics.com/rev-crossover-products/ultraplanetary/up-gearbox>

REV Ultra 90 gearbox docs

<https://docs.revrobotics.com/rev-crossover-products/ultraplanetary/ultra-90>

REV DUO Starter Bot build guide (PDF) - the Mk-1 robot

https://www.revrobotics.com/content/technical-resources/DUO/FTC-Kickoff-Concepts/2026-27/2026-27_REV_DUO_FTC_Starter_Bot-Build_Guide_1567.pdf

REV Starter Bot docs - BOM, TeleOp/Auto code, upgrades

<https://docs.revrobotics.com/ftc-kickoff-concepts>

REV Starter Bot CAD - public Onshape document

<https://cad.onshape.com/documents/9cb9eb90e973fd37d10ee178/w/b/9c02b0f4e09998f08becbd2/e/a55c018c62564f3ea8812a47>

BIOBUZZ Competition Manual & game materials

<https://ftc-resources.firstinspires.org/ftc/game>

REV Hardware Client guide (Windows)

<https://docs.revrobotics.com/rev-hardware-client>

FTC Docs - programming reference

https://ftc-docs.firstinspires.org/en/latest/programming_resources/index.html

Game Manual Zero (gm0) - start here

<https://gm0.org/en/latest/docs/start-here.html>

FTC SIM - browser robot simulator

<https://ftcsim.org/>

Virtual Robot Simulator

<https://www.vrobotsim.online/homepage.html>

FTC Community Forum

<https://ftc-community.firstinspires.org/>

Onshape for FIRST teams (free CAD)

<https://www.onshape.com/en/education/first-robotics#ftc>

Learn Java for FTC - free PDF, optional reference

<https://github.com/alan412/LearnJavaForFTC>

Programming board 3D model - the board this team built

<https://www.thingiverse.com/thing:6864893>

Running the meeting

Coach page — Team 22257 · Tue 4:00–6:00 PM & Fri 12:00–3:00 PM, Room 200

Two meeting shapes

DAY	TASK TIME	HOW IT RUNS
Tuesday 4:00–6:00	100 min	10 min orientation → one card per crew → 10 min cleanup. A card is built to fill exactly this.
Friday 12:00–3:00	160 min	10 min orientation → one card, then 60 more minutes → 10 min cleanup. Spend the extra hour finishing, on the card's "If you finish early" box, or starting the next card in that crew's chain. On Fridays late in the season, use it to run a Skill Builders challenge as a whole team.

Where we are — and the road to three qualifiers

Meetings 1–4 (Sept 15–25) took the robot from an unopened kit to a drivable mecanum base running Fast Lane. That phase is finished. Everything below begins at the first meeting back from fall break, **Tue Oct 13**.

QUALIFIER	DATE	WHAT WE BRING
Roadrunner Gilbert Christian School, Gilbert	Sat Nov 14	Mk-1 — the REV DUO Starter Bot superstructure built to REV's guide, mounted on <i>our</i> mecanum chassis. The goal is a robot that scores, on time, not a clever one.
Saguaro Blossom Microchip, Chandler	Sat Jan 23	Mk-2 — our own design, prototyped in parallel from October and built over the winter on what Roadrunner taught us.
Copper American Leadership Academy, Queen Creek	Sat Feb 6	Mk-2 refined — two weeks to fix whatever January exposed. No new mechanisms this late.

Ten meetings to Roadrunner

Five weeks back, two meetings each. **The key scheduling fact: four of the five Starter Bot assemblies are bench work** — launcher floor, intake, flywheel & feeder, and pollen pusher all get built off the robot. Four crews can build them at once in week one, which is the only reason ten meetings is enough.

WEEK	MK-1 TRACK — THE ROBOT THAT COMPETES	MK-2 & MEDIA TRACK — RUNS ALONGSIDE, EVERY MEETING
1 Oct 13, 16	► BA-11 Parts check & cut list — <i>everything else waits on this</i> . Then BA-12 , BA-13 , BA-14 , BA-15 start in parallel on the bench.	CD-9 Fit the Starter Bot to our chassis in CAD — do this in week one, the build crew needs its numbers. MF-2/MF-3 sponsor letters. EP-13 build the field elements the day the kit lands.
2 Oct 20, 23	Finish the four subassemblies. ► BA-16 Mount them on the mecanum chassis — the part REV has no guide for.	CD-3/CD-4 intake and outtake concepts, BA-7 cardboard prototype. EP-11 portfolio started.
3 Oct 27, 30	► BA-17 Expansion Hub → PR-18 Configuration → PR-19 Starter Bot TeleOp on mecanum. It drives and shoots by Oct 30 .	CD-6/CD-7 printed parts, MG-2/MG-3 give-aways for the event, TS-5 press the shirts.
4 Nov 3, 6	PR-20 Flywheel tuning, PR-21 Autonomous, EP-9 Drive team practice. Most of the points live in this week.	EP-11 portfolio submitted, EP-8 scouting sheet built and tested.
5 Nov 10, 13	► BA-18 Inspection dry run — a week early, on purpose. Then drive practice only. No new mechanisms .	EP-10 Pit setup & run sheet, pack the tote, rehearse the interview.

If you fall behind. The one date that cannot move is **BA-16 finished by Oct 23**. Everything after it is programming and practice, and programming can be compressed; building cannot. If week two slips, drop the pollen pusher (**BA-15**) — the robot scores without it. If week three slips, go to Roadrunner with TeleOp only and no autonomous. **A robot that drives and shoots badly beats a robot that stays in the pit.**

After Roadrunner. Run **EP-12** at the very first meeting back, while everyone still remembers. It turns what the team just watched into the ranked list that decides what Mk-2 actually is. Then there are roughly fourteen meetings before Saguaro Blossom on Jan 23 — minus Thanksgiving and winter break — which is enough to design and build one good mechanism, and not enough to build three.

Hardware & the second robot

Coach page — what can run at the same time

The hardware column — how to run two tracks at once

Every card is tagged with the hardware it needs. This is the constraint that decides what can run in parallel: **46 of the 79 cards need no hardware at all**, so there is always something for a crew to pick up.

TAG	CARDS	WHAT IT MEANS FOR PLANNING
No hardware	46	Runs anywhere, any number of crews at once. Planning, all nine CAD cards , merch, shirts, marketing, PR-9 (simulator), BA-1/BA-6, and — the reason the Starter Bot fits in ten meetings — the four bench subassemblies BA-12 to BA-15 . <i>CD-6 and CD-7 need printer access — a shared resource, but not one that competes for the robot.</i>
Programming board	4	Needs the 3D-printed board. BA-9 builds it; PR-10, PR-11 and PR-12 are the sensor cards only the board can teach — touch, color and potentiometer are not on the StarterBot base at all.
Board or robot	3	Either one works. Run these on the board while the build crew still has the robot apart.
Robot / Robot + field	26	The scarce resource — one crew at a time, unless the class robot is both present and has its own hub. “Robot + field” also needs EP-6 done first.

Hub accounting — and the one that will stop the season. A Control Hub has four motor ports. Mecanum uses all four. The Starter Bot’s intake and flywheel are motors **five and six**, so **Mk-1 cannot work without an Expansion Hub** (REV-31-1153, about \$275, ships with the XT30 extension and the 3-pin communication cable it needs). There is no software way around this and no way to borrow a Control Hub instead — a Control Hub cannot act as an Expansion Hub. **We own one from last season and it is marked flaky.** BA-17 now tests it before trusting it: firmware update in the REV Hardware Client first (stale firmware is the usual cause), then every port exercised and every cable wiggled under load. We only need two good ports, so one dead port is survivable; a hub that drops the whole link is not — and an intermittent hub at an event is indistinguishable from a software bug. If it drops even once, order the replacement and keep the old one as the bench spare.

The Starter Bot parts. REV’s guide assumes a complete **Starter Kit V3.1** on the bench. We have the parts from an earlier season’s kit plus the team’s own library, so most of the superstructure should already be on the shelf — **BA-11 is the inventory that proves it**, and it is the first card back from break for exactly that reason. Worth checking early because it is consumable: **15mm extrusion** (the guide eats 420mm x9 and 225mm x5), **polycarbonate sheet** (REV-41-7537), **8 intake flaps**, **4x 90mm grip wheels**, **4x 1in soft grip wheels**, **4x 90T and 1x 60T gears**, **#25 chain with 10T/20T sprockets**, **2x Smart Robot Servo V2**, a **Core Hex motor**, one spare **UltraPlanetary + HD Hex** for the flywheel, and M3 hardware in bulk — the guide pre-loads 195 screws and 195 nyloc nuts before step one. Anything short gets ordered the day BA-11 finds it.

When the Herobotix class robot is available

It is shared, so plan every meeting around one robot and treat the second as a bonus. It runs the **same REV Control Hub**, so OpModes port over — but a configuration saved on one hub does not exist on the other, so it needs its own config built (PR-2) before anything runs. **First check it still has a hub** (see hub accounting above). On the meetings it shows up ready, these are the cards worth reaching for:

CARD	WHY A SECOND ROBOT CHANGES IT
PR-5 Driver Tuning	Two drivers practice at once instead of taking turns. Doubles the reps in the same 100 minutes, and the penalty counts become comparable between drivers.
EP-6 Field — Fast Lane	The Skill Builders deck suggests a wider track for multiple robots and a Grand Prix format. That needs two robots to exist.
PR-6 / PR-7 Autonomous	One robot runs while the other crew edits code, instead of both crews queuing for the same robot between test runs.
PR-13 Bench to Robot	Port the same bench behavior onto both robots. Whatever breaks on the second one is the part the students did not actually understand.
MF-6 Demo Event	Two robots make a far better demo, and lets more students drive in front of an audience.

Team Identity: Name, Look & Motto

No hardware

Start here

3-5 people

100 min

WHY THIS MATTERS

Everything downstream needs a name and a look — shirts, buttons, sponsor letters, the website. Until this is done, the merch and marketing tracks are blocked. Do it first, do it once.

YOU WILL FINISH WITH

One team name, one backup, a written identity brief, and three rough logo directions on paper.

WHAT YOU NEED

- Whiteboard or big paper
- Markers
- Sticky notes
- A phone or laptop to search names

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

MF-2, MF-4, MF-5, MG-1, MG-2, MG-4, MG-6, TS-1

IF YOU FINISH EARLY

- Write a 15-second team introduction that any member could say to a judge or a sponsor without notes.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Get a big writing surface and hand everyone a marker.
- ☐ Look up the other Herobotix team names so you do not pick one that is taken.
- ☐ Write the season name (BIOBUZZ) and the scoring element (Pollen) on the board — both are good idea fuel.

15-45 min

- ☐ SILENT round: everyone writes 5 name ideas on sticky notes. No talking, no judging. 5 minutes.
- ☐ Post every note on the board. Read them all out loud.
- ☐ Group similar ideas together. Still no criticism — that comes next.

45-65 min

- ☐ Score each idea out loud: Is it easy to say? Does it fit on a shirt? Can you draw it? Is it already taken (search it)?
- ☐ Cut the board down to 3 finalists.

65-85 min

- ☐ Vote. Everyone gets one vote. Pick a winner and a backup.
- ☐ Write both on the board and photograph it.

85-100 min

- ☐ Write the identity brief — 3 sentences: who we are, what we are about, what our look is.
- ☐ Name 2-3 team colors and one shape or mascot idea.
- ☐ Sketch 3 rough logo directions. Rough is fine. Tape them up.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Name and backup written down and photographed
- ☐ 3-sentence identity brief in the engineering notebook
- ☐ 3 logo sketches taped to the wall

HAND OFF TO

Hand the identity brief and logo sketches to whoever takes TS-1 (shirt design) and MG-2 (give-away art).

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Every decision this season traces back to the Competition Manual. If nobody on the team has actually read the scoring section, you will build the wrong robot and find out in January.

YOU WILL FINISH WITH

A one-page game summary poster on the wall that anyone can point at mid-argument.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-10 min

- ☐ Open the Competition Manual and find the scoring section.
- ☐ Split the scoring rules among the crew — one person per section.

10-40 min

- ☐ Read your section. Write down every way to score, what it is worth, and when it is allowed (autonomous / driver-controlled / endgame).
- ☐ Write down anything you do not understand — you will ask the coach.

40-70 min

- ☐ Build the poster: a sketch of the field, a scoring table, and a match timeline bar showing each phase and how many seconds it lasts.

70-90 min

- ☐ Add a GOTCHAS box: the top 5 penalties and what each one costs.

90-100 min

- ☐ Present the poster to the rest of the team in 3 minutes. Hang it where the robot gets built.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ Poster is on the wall where the build happens
- ☐ Scoring table matches the Competition Manual exactly
- ☐ Anyone on the team can name the three match phases and their lengths

WHAT YOU NEED

- Competition Manual (printed or on a laptop)
- Poster paper
- Markers
- Ruler

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

CD-4, EP-3, EP-8

IF YOU FINISH EARLY

- Add a points-per-second column. Which scoring method is actually worth chasing when the clock is running?

HAND OFF TO

The build and programming crews will use this poster constantly. Tell them it exists.

Who worked on this: _____ Date: _____

Robot Requirements List

WHY THIS MATTERS

'Build a robot' is not a plan. A requirements list turns the rulebook into specific things you can build and then actually check off.

YOU WILL FINISH WITH

A numbered requirements list, each labeled MUST / SHOULD / NICE, each with a way to test it.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Pull up the game poster and the robot/inspection rules in the Competition Manual.

15-45 min

- ☐ Write requirements as testable sentences. Good: 'Robot fits inside the legal starting size.' 'Robot picks up one Pollen off the floor.' Bad: 'Robot is fast.'
- ☐ Aim for 15-25 requirements. Number them.

45-65 min

- ☐ Label each one MUST, SHOULD, or NICE. Argue about it — the argument is the valuable part.
- ☐ MUST means the robot is illegal or useless without it.

65-85 min

- ☐ Add a HOW WE TEST IT column to every requirement. If you cannot describe the test, it is not a requirement yet — rewrite it.

85-100 min

- ☐ Post it next to the game poster. Walk the build crew through it.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ List is numbered and every item is labeled MUST / SHOULD / NICE
- ☐ Every MUST has a written test
- ☐ Posted next to the game poster

WHY THIS MATTERS	YOU WILL FINISH WITH	WHAT YOU NEED
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- The EP-2 game poster
- Competition Manual (inspection + robot rules section)
- Laptop or large paper

BEFORE YOU START

EP-2

THIS UNLOCKS

EP-4

IF YOU FINISH EARLY

- Add a STATUS column (Not started / In progress / Passes). This list is now your build tracker for the whole season.

HAND OFF TO

This is the scorecard for EP-4 concept selection and for every build task.

Who worked on this: _____ Date: _____

Design Sketches & Pick a Concept

No hardware

Some experience

3-4 people

100 min

WHY THIS MATTERS

The first idea is rarely the best one, and the loudest person's idea is not automatically right. Sketching three and scoring them takes an hour and can save weeks of rebuilding.

YOU WILL FINISH WITH

Three sketched concepts for one mechanism, scored against your requirements, one chosen with a written reason.

WHAT YOU NEED

- Paper and pencils (one sheet each)
- The EP-3 requirements list
- Whiteboard

BEFORE YOU START

EP-3

THIS UNLOCKS

BA-5, BA-7, CD-3

IF YOU FINISH EARLY

- Sketch the backup plan and write down what would have to go wrong before you switch to it.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-10 min

- ☐ Pick ONE mechanism to design today: the intake, the scoring/outtake, or how Pollen gets stored on the robot. One. Not all three.

10-40 min

- ☐ Everyone sketches independently. No looking at each other's paper. Label the parts. Draw at least 2 views (side and top).

40-55 min

- ☐ Pin up all sketches. Each person explains theirs in 90 seconds. Listeners ask questions, not criticisms.

55-80 min

- ☐ Build a decision matrix on the whiteboard. Rows = concepts. Columns = 'Can we build it with parts we own?', 'How many meetings to build?', 'How likely to break?', 'Points it earns'.
- ☐ Score every box 1 to 5. Total the columns.

80-95 min

- ☐ Pick the winner. Write one paragraph explaining WHY — including what you gave up by not picking the others.

95-100 min

- ☐ Photograph every sketch and the matrix into the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ 3 or more labeled sketches with 2 views each
- ☐ Decision matrix filled in and totaled
- ☐ A written paragraph explaining the choice and the trade-off

HAND OFF TO

The chosen concept goes to BA-7 (prototype) and into the engineering notebook.

Who worked on this: _____ Date: _____

Set Up the Engineering Notebook

No hardware

Start here

2-3 people

100 min

WHY THIS MATTERS

Judges read the notebook. More importantly, it is how you remember in February why you made a choice back in October. A team without a notebook resolves the same problem three times.

YOU WILL FINISH WITH

A working notebook with sections, a reusable daily template, and today's entry already written.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Ask the coach what format the team is using: physical binder, shared doc, or both. Set it up.

15-35 min

- ☐ Build the sections: Team & Roles / Meeting Log / Design (one per mechanism) / Programming / Outreach / Business & Budget.

35-55 min

- ☐ Make the one-page daily entry template: Date | Who was here | What we set out to do | What we actually did | What we learned | What is next.
- ☐ Print 20 copies, or duplicate the page in the shared doc 20 times.

55-80 min

- ☐ Back-fill an entry for every meeting so far, using photos and memory. Do not skip this — it only gets harder every week.

80-95 min

- ☐ Write today's entry properly, as the example everyone else will copy.

95-100 min

- ☐ Write the name of next meeting's notebook owner on the whiteboard.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ All sections exist
- ☐ Daily template exists and is stocked
- ☐ Every past meeting has an entry
- ☐ Today's entry is written
- ☐ Next owner named

WHAT YOU NEED

- Binder and dividers, or a shared doc (ask the coach which the team is using)
- Printer access
- Photos from past meetings

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

EP-11

IF YOU FINISH EARLY

- Add at least one photo to every back-filled entry.

HAND OFF TO

Notebook duty rotates. Whoever is named owns capturing the next meeting.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

You cannot practice driving or test an autonomous routine without a marked field. This is also the setup for the Fast Lane and Wayfinder Skill Builders challenges.

YOU WILL FINISH WITH

A taped practice area with correct, measured dimensions, ready for driving challenges.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ ASK THE COACH before taping anything — confirm the floor is OK and only painter's or gaffer's tape is used.
- ☐ Measure the room. Find the largest clear rectangle you can use.

15-30 min

- ☐ Mark the four corners. Measure both diagonals — if they are equal, your rectangle is square. Adjust until they match.

30-60 min

- ☐ Lay the outer boundary tape. Then lay a Fast Lane track inside it, at least 2 ft (0.61 m) wide everywhere.
- ☐ Track shape is your choice — see the Fast Lane slides for oval and figure-8 options.

60-80 min

- ☐ Mark a Start/Finish line, an Alliance Area off the field where drivers stand, and at least 4 checkpoints for Wayfinder.

80-95 min

- ☐ Drive a robot around it (or walk the path holding a 18 in box). Fix any corner that is too tight.

95-100 min

- ☐ Photograph the layout and write the measurements onto the photo.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Diagonals measured equal
- ☐ Track is at least 2 ft wide the whole way around
- ☐ Start/Finish, Alliance Area, and 4+ checkpoints marked
- ☐ Measured photo in the notebook

WHAT YOU NEED

- Painter's or gaffer's tape in a high-contrast color
- Tape measure (25 ft)
- The Fast Lane and Wayfinder challenge slides

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

PR-17, PR-5, PR-6

IF YOU FINISH EARLY

- Build a simple removable border out of PVC or 1x4 lumber so the robot cannot run off the field.

HAND OFF TO

Tell the programming crew the field is up — PR-5, PR-6 and PR-7 all need it.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Nobody touches a tool until they are checked off. It is also the fastest way for a brand-new member to go from watching to being useful.

YOU WILL FINISH WITH

Every member signed off on the safety rules and on each hand tool they are cleared to use.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Coach walks the whole group through the space: exits, first aid kit, fire extinguisher, where safety glasses live, where batteries live.

15-35 min

- ☐ As a group, write the team's OWN safety rules in your own words — 8 to 12 of them. Post them on the wall.

35-65 min

- ☐ HAND TOOL stations — hex drivers, wrenches, cutters, files. One person demonstrates, then EVERY member does one supervised repetition.
- ☐ These are the tools the build cards actually call for, so everyone gets cleared on them today.

65-85 min

- ☐ POWER TOOLS are a different thing and today you do not run them. Walk to the chop saw, scroll saw, belt sander and drill press as a group.
- ☐ The coach names each one, shows where its switch and its stop are, and explains that it stays off-limits until you are individually signed off by a mentor.
- ☐ Read the Power Tool Check-Off poster out loud together — especially the line that a signature never means you may use it alone.

85-100 min

- ☐ Fill in the hand-tool grid and post both sheets where the tools live.
- ☐ Write down who still needs clearing. Power tool sign-offs happen later, two or three students at a time with a mentor — never a whole group in one session.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ Team-written safety rules posted
- ☐ Hand-tool grid filled in for everyone present and posted
- ☐ Power Tool Check-Off poster up, read aloud, and understood — nobody ran a power tool today
- ☐ Safety glasses counted and located
- ☐ A written list of who still needs clearing

WHAT YOU NEED

- Safety glasses for everyone
- The team's hand tools
- The printed hand-tool check-off grid
- The printed Power Tool Check-Off poster
- Coach or mentor to lead

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

BA-11, BA-2, BA-4, BA-9, EP-13

IF YOU FINISH EARLY

- Make a laminated one-page safety card to live in the pit toolkit at competitions.

HAND OFF TO

The build tasks assume the hand-tool clearing is done. Power tool sign-offs are a separate, ongoing thing a mentor does in small groups — keep both grids current as new members join.

Who worked on this: _____ Date: _____

Scouting & Alliance Strategy

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

At a qualifier you play a handful of matches with a random partner, and then the top teams pick alliances. Teams that know what the other robots can actually do pick better, play better, and get picked. Scouting is how a small team punches above its weight.

YOU WILL FINISH WITH

A printed scouting sheet, a practice round filled in from match video, and a one-page card for our alliance partner.

WHAT YOU NEED

- Competition Manual
- Clipboards and a printer
- Laptop with match video from a past event
- Timer

BEFORE YOU START

EP-2

THIS UNLOCKS

EP-12

IF YOU FINISH EARLY

- Add the column nobody else tracks — how long the robot takes between scores. Cycle time predicts winners better than raw score does.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Read the scoring section of the Competition Manual out loud. Write every way to score Pollen and Nectar on the board with its point value.

20-45 min

- ☐ Design the sheet. One row per match; columns only for things you can count from the stands in two seconds. If you cannot count it that fast, it does not belong on the sheet.

45-70 min

- ☐ Test it. Play match video and have two students fill the same sheet live. Wherever the two sheets disagree, that column is badly worded — fix it.

70-90 min

- ☐ Write the alliance partner card: what our robot does, what it needs from a partner, where we start in auto, and what we will NOT do. One page, in a sheet protector.

90-100 min

- ☐ Decide who scouts on event day and who backs them up. Put the names on the run sheet.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Scouting sheet printed, 10+ copies on clipboards
- ☐ Two students agree within 10% scouting the same video
- ☐ Alliance partner card written
- ☐ Scouts and backups named

HAND OFF TO

The partner card goes in the pit binder (EP-10). The scouting data feeds the debrief in EP-12.

Who worked on this: _____ Date: _____

Drive Team: Roles, Calls & Practice

Robot + field

Some experience

3-4 people

100 min

WHY THIS MATTERS

A match is two and a half minutes long and the robot only does what the drive team makes it do. Most rookie teams lose points to confusion, not to hardware. Three defined jobs and a few rehearsed calls are worth more than another motor.

YOU WILL FINISH WITH

A named drive team with defined roles, a set of calls everyone says the same way, and 10 timed practice matches logged.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Assign three jobs: Driver drives, Operator runs mechanisms, Coach watches the field and the clock and talks. Anyone can try out — the job goes to whoever is steadiest, not loudest.

15-30 min

- ☐ Write the calls. Short words everyone says identically: loaded, shooting, jammed, thirty seconds, reset. Tape the list to the Driver Hub cart.

30-80 min

- ☐ Run 10 timed matches — 30 seconds auto (sit still if auto is not ready), then driver control. Log Pollen scored and anything that broke. Rotate one person each round so every job has a backup.

80-95 min

- ☐ Read the log. Where did the time actually go? Rank the three biggest time-wasters.

95-100 min

- ☐ Fix the easiest one right now. It is usually where the robot parks to reload.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Three roles assigned, each with a named backup
- ☐ Call list written and posted at the driver station
- ☐ 10 matches logged with scores
- ☐ Top three time-wasters written down

WHAT YOU NEED

- Robot
- Taped practice field (EP-6)
- Driver Hub and both gamepads
- Timer
- All the Pollen you have
- Notebook

BEFORE YOU START

PR-19

THIS UNLOCKS

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IF YOU FINISH EARLY

- Run one match with the Coach facing away from the field, calling only on what the Driver says out loud. It exposes every gap in the calls.

HAND OFF TO

Give the worst time-waster to the Mk-2 crew and to EP-12 — that is a design requirement, not a driving problem.

Who worked on this: _____ Date: _____

Pit Setup & Event Day Run Sheet

No hardware

Start here

2-3 people

100 min

WHY THIS MATTERS

A qualifier starts early and runs long. Everything you forget at home you will spend the day borrowing. A packed pit and a run sheet with names on it turn a stressful day into a normal one.

YOU WILL FINISH WITH

A packed pit tote, a printed run sheet naming who does what, and a pit that goes up in under 15 minutes.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Read the event page and put the real times on the board: doors open, inspection, opening ceremony, first match, awards. Work backwards to a departure time from Mesa, then add 30 minutes.

20-50 min

- ☐ Pack the tote against a written list: full tool set, spare hardware, charged batteries and charger, laptop and cables, gaffer tape, zip ties, spare gamepad, the build guide, safety glasses for everyone, snacks and water.

50-75 min

- ☐ Build the pit display: team sign, the portfolio, give-aways from MG-3, and somewhere to set the robot down that is not the floor.

75-95 min

- ☐ Write the run sheet — every hour of the day, with names. Who drives, who scouts, who fixes, who talks to judges, who hands out give-aways. Print one per student and one for parents.

95-100 min

- ☐ Dry-run the pit setup in the shop and time it.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Tote packed against the written list
- ☐ Pit display set up in under 15 minutes in the shop
- ☐ Run sheet printed with every student named
- ☐ Departure time agreed and sent to families

WHAT YOU NEED

- Robot cart and pit kit (BA-8)
- The event's team information packet
- Printer
- Big plastic tote
- Tape and zip ties

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Add a laminated 'if the robot breaks' card — the three most likely failures and which spare fixes each one.

HAND OFF TO

The run sheet goes to the coach and to every family. The tote stays packed until the event.

Who worked on this: _____ Date: _____

Judging: Portfolio & the Five-Minute Interview

No hardware

Challenge

3-4 people

100 min

WHY THIS MATTERS

Much of what a team can win at a qualifier is decided by judges, not by matches. They read the Engineering Portfolio and they talk to you for a few minutes. A rookie team that can explain its own decisions beats a fancier robot that cannot.

YOU WILL FINISH WITH

A submitted Engineering Portfolio and an interview that every student takes part in.

WHAT YOU NEED

- The engineering notebook (EP-5)
- Photos from every build card
- This season's portfolio requirements and judging rubric
- Laptop and printer

BEFORE YOU START

EP-5

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Practise the question nobody wants: 'What would you change about this robot?' A team that answers that one well sounds like engineers.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Download this season's portfolio requirements and the judging rubric. Read the page limit and the deadline out loud, then write the deadline on the wall calendar.

15-55 min

- ☐ Assemble it from what you already have: the identity brief (EP-1), the requirements list (EP-3), the sketches (EP-4), build photos, and the decision to run the REV Starter Bot first and design our own second. Judges reward decisions with reasons, not pretty pictures.

55-80 min

- ☐ Write the interview. Everyone speaks. Split it four ways: who we are, what the robot does, one thing that went wrong and what we did about it, what is next. Under five minutes total.

80-95 min

- ☐ Rehearse twice in front of the coach, who asks one hard question each time.

95-100 min

- ☐ Check the submission method and deadline again, then submit or schedule the submission.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Portfolio inside the page limit and submitted before the deadline
- ☐ Every student has a speaking part
- ☐ Rehearsed twice with questions
- ☐ Deadline on the wall calendar

HAND OFF TO

Print two copies for the pit binder (EP-10) — judges sometimes visit the pit.

Who worked on this: _____ Date: _____

Post-Event Debrief: What Mk-2 Must Fix

No hardware

Some experience

whole team people

100 min

WHY THIS MATTERS

The two weeks after a qualifier are the most valuable of the season — you have just watched thirty robots and you know exactly what yours cannot do. Teams that write it down before they forget build a better second robot. Teams that do not repeat November in January.

YOU WILL FINISH WITH

A ranked list of changes for Mk-2, agreed by the whole team, with an owner and a task card for the top three.

WHAT YOU NEED

- Scouting data (EP-8)
- Drive team log (EP-9)
- Match results
- Whiteboard
- Engineering notebook

BEFORE YOU START

EP-8

THIS UNLOCKS

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IF YOU FINISH EARLY

- Send a thank-you to the volunteers and to any team that lent you a part. It takes ten minutes and it is how small teams get help next time.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Facts first, no opinions. Put the numbers on the board: matches played, average score, Pollen scored per match, how many times the robot stopped working.

20-45 min

- ☐ Round the room. Everyone names one thing that worked and one thing that did not. Write all of it down. No arguing yet.

45-70 min

- ☐ Group the failures: too slow, could not reach, broke, could not be driven, could not be programmed. Now argue — and rank the top five by how many points each one cost.

70-90 min

- ☐ For the top three, name an owner and pick the card that fixes it. If no card fits, write what a new card would need to cover and tell the coach.

90-100 min

- ☐ Write the debrief into the notebook with the date and the ranked list. Judges at the next event will ask about it.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Numbers on the board and in the notebook
- ☐ Top five ranked by points cost
- ☐ Top three have an owner and a card
- ☐ Dated notebook entry

HAND OFF TO

The ranked list drives every CAD and building card between now and the next qualifier.

Who worked on this: _____ Date: _____

Build the BIOBUZZ Field Elements

No hardware

Some experience

4 people

160 min

WHY THIS MATTERS

The Flowers and Hives are what the robot scores into, and they arrive as flat plastic, tubes and hardware bags. FIRST's assembly guide is 42 pages and strict about orientation — a Hive built slightly twisted tips differently from the real one, and every shot tuned against it is tuned wrong. No robot needed: this runs while the build crew has it.

YOU WILL FINISH WITH

Four Flowers, four Cells, two Hives and the Frame assembled to FIRST's guide, checked flat, and stored safely.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Unbox onto one table and check every bag against the parts list in section 2. Read section 3 out loud — 'tight' means something specific for each fastener type, and cable-tie tails are cut flush, never left sharp.
- ☐ Split into two pairs: Flowers and Cells.

20-80 min

- ☐ Flowers pair: section 5, four times. Watch the lettered rings (A, X, B, C) and which way the top backstop faces the wall bracket.
- ☐ Cells pair: section 6, four times — two red, two blue. Place the AprilTag stickers exactly by the reference holes; autonomous code will read them.

80-120 min

- ☐ Together: section 7, two Hives. The pivot bearings are a tight fit — press them in with a clamp or tap gently, never force. Leave the screws loose until you have checked the Hive is flat and not twisted, then tighten.

120-150 min

- ☐ Section 8: the Frame that carries the Hives. Feet A and B are not interchangeable. Then fold the two scoring element trays (section 9).

150-160 min

- ☐ Store the Hives with nothing stacked on them (section 10). Count Pollen and Nectar into labelled bins. Photograph everything for the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ Every part checked against the parts list
- ☐ Four Flowers, four Cells, two Hives and the Frame assembled
- ☐ Hives checked flat and untwisted before final tightening
- ☐ Every cable-tie tail flush-cut
- ☐ Hives stored with nothing on top of them

WHAT YOU NEED

- FIRST's Initial Field Element Assembly Guide
- The AndyMark BIOBUZZ kit
- 5/32 hex driver, 3/8 wrench and nut driver, PH2 driver, drill
- Flush cutters, utility knife, tape measure
- Clamp or mallet for the Hive bearings
- Safety gloves

LINKS

Clickable on the Coaches Corner site

• FIRST Initial Field Element Assembly Guide

ftc-resources.firstinspires.org/ftc/field/initialfieldguide

BEFORE YOU START

EP-7

THIS UNLOCKS

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IF YOU FINISH EARLY

- Mount a Hive on the Frame, tune its ballast washers with the Event Field Setup Guide, then drop Pollen in until it tips. That tipping point is the target PR-20 is shooting at.

HAND OFF TO

Tell the programming crew the Hive exists — PR-19, PR-20 and PR-21 all end with shots at it.

Who worked on this: _____ Date: _____

Onshape Setup & Your First Part

No hardware

Start here

2-4 people

100 min

WHY THIS MATTERS

Onshape is free for FIRST teams, runs in a browser on any computer including a Chromebook, and it is what professional engineers actually use. Everything the team designs from here lives in it.

YOU WILL FINISH WITH

Everyone has an account in the team workspace and has modeled a real part with exact dimensions.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Everyone creates a free Onshape account through the FIRST education link. The coach approves the signups — students should not be creating accounts unsupervised.
- ☐ Join the team's shared Onshape workspace so your work is visible to everyone.

20-45 min

- ☐ Interface tour together: the Part Studio, a sketch, the extrude tool, and the feature tree down the left side.
- ☐ Follow one intro tutorial as a group before anyone works alone.

45-80 min

- ☐ Everyone models the SAME part so you can compare: a flat plate 100mm x 60mm x 6mm with two 5mm holes 70mm apart.
- ☐ Sketch it on a plane, dimension it properly (do not eyeball it), extrude it, then cut the holes.
- ☐ The loop you are learning is sketch, dimension, extrude. Everything else is a variation of it.

80-95 min

- ☐ Now change the plate's length from 100mm to 130mm and watch the whole part update. That is what makes CAD different from drawing a picture.

95-100 min

- ☐ Share your part with the coach. Write down the three tools you used most.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Account created and joined to the team workspace
- ☐ Plate modeled to the exact dimensions given, not eyeballed
- ☐ A dimension changed and the model updated correctly
- ☐ Part shared with the coach

WHAT YOU NEED

- One computer per person
- The Onshape for FIRST link on the resource shelf
- Coach present to approve account signups
- A ruler for reference

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

CD-2

IF YOU FINISH EARLY

- Model the same plate again from scratch without looking at notes. In CAD, speed is entirely repetition.

HAND OFF TO

CD-2 is the next step. Tell the team the workspace exists and who is in it.

Who worked on this: _____ Date: _____

Measure & Model a Real Robot Part

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

CAD is only useful if it matches reality. Modeling something you can hold, then checking your model against it with calipers, is how you learn to trust your own drawings — and how you catch that you have been guessing.

YOU WILL FINISH WITH

An accurate Onshape model of a real part from the kit, verified against the physical part.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Pick a real part with a few holes in it. Do not pick the most complicated thing on the shelf for your first one.
- ☐ Get calipers and learn to read them — ask a mentor if you have not used them.

15-40 min

- ☐ Measure it properly and write EVERY dimension on paper before you model anything.
- ☐ Measure each dimension twice. If the two readings disagree, measure a third time and find out why.

40-80 min

- ☐ Model it in Onshape from your written measurements — not from looking at the part on the table.

80-95 min

- ☐ Check your work: compare the model's dimensions against the physical part with calipers. Anything more than about 0.5mm off, find out why before you move on.

95-100 min

- ☐ Save it into the team's parts library with a name someone else would recognize.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ A written measurement sheet made BEFORE modeling started
- ☐ Model built from the sheet
- ☐ Every dimension verified against the real part within about 0.5mm
- ☐ Saved to the team library with a clear name

WHAT YOU NEED

- Onshape account (CD-1)
- Calipers (ask the coach)
- A real part: a C-channel offcut, a bracket, a spacer, a wheel hub
- Paper for the measurement sheet

BEFORE YOU START

CD-1

THIS UNLOCKS

CD-3, CD-4, CD-5, CD-6, CD-8

IF YOU FINISH EARLY

- Model a second part and mate the two together in an Assembly. That is the skill CD-5 is built on.

HAND OFF TO

The team parts library is what makes CD-3, CD-4 and CD-5 fast. Keep adding to it.

Who worked on this: _____ Date: _____

Design a Pollen Intake in CAD

WHY THIS MATTERS

The Mecanum Drivetrain Kit is a drive base and nothing else, so the intake is entirely your team's design. Working it out in CAD first means you find the problems before anyone cuts metal or burns three hours of filament.

YOU WILL FINISH WITH

A CAD model of an intake concept, sized against a real Pollen and the real mounting space.

WHAT YOU NEED

- Onshape (CD-2)
- The concept chosen in EP-4
- Calipers
- A real Pollen to measure
- Access to the robot to measure the mounting space

BEFORE YOU START

CD-2, EP-4

THIS UNLOCKS

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IF YOU FINISH EARLY

- Model a second, genuinely different concept and put both in front of the team. Two options make a better decision than one.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Get real numbers before you draw anything: measure a Pollen with calipers, and measure the space on the robot where this has to mount. Write both down.
- ☐ Review the concept the team chose in EP-4 — you are modeling that, not starting a new argument.

20-50 min

- ☐ Model the working part first — the thing that actually touches the Pollen. Keep it simple; simple things get built.

50-75 min

- ☐ Model the mounting. How does it attach to the C-channel? Measure the channel's real hole spacing and match it exactly, or it will not bolt on.

75-95 min

- ☐ Sanity-check in CAD: does a Pollen-sized object fit through the opening? Is anything thinner than 3mm, which will snap? Does it stick out past the robot's legal starting size?

95-100 min

- ☐ Share it with the build crew and write down honestly what you are unsure about.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Pollen and mounting space both measured and written down
- ☐ Intake modeled with mounting holes that match the real channel spacing
- ☐ Clearance checked against a Pollen-sized object in CAD
- ☐ Shared with the build crew with open questions noted

HAND OFF TO

BA-5 builds it. CD-6 can print parts of it.

Who worked on this: _____ Date: _____

Design a Pollen Outtake in CAD

No hardware

Challenge

2-3 people

100 min

WHY THIS MATTERS

Getting Pollen in is one problem; putting it where you want it is a different one. An outtake has to release on command, aim, and not jam — and that geometry is far easier to try in CAD than in aluminum.

YOU WILL FINISH WITH

A CAD model of an outtake concept with the Pollen's path through it checked.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Look at the EP-2 game poster: where does Pollen actually have to go, and how high? Write those numbers down — they are the requirements.

20-45 min

- ☐ Sketch the Pollen's path through the mechanism ON PAPER first: where it enters, where it waits, where it exits, and what pushes it out.

45-80 min

- ☐ Model it. Model a simple Pollen shape too, so you can place it inside your mechanism and actually see whether the path works.

80-95 min

- ☐ Check it: does anything block the exit? Is the release repeatable, or does it depend on the Pollen sitting perfectly? Mark where it would jam.

95-100 min

- ☐ Share with the build crew and note what you are unsure about.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Scoring target and height recorded from the game poster
- ☐ Pollen path sketched on paper before modeling
- ☐ Mechanism modeled with a Pollen positioned inside it
- ☐ Likely jam points identified and noted

WHAT YOU NEED

- Onshape (CD-2)
- The EP-2 game poster for scoring targets and heights
- Calipers and a real Pollen
- Paper for the path sketch

BEFORE YOU START

CD-2, EP-2

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Model it in two positions, loaded and released, so the team can see the motion instead of imagining it.

HAND OFF TO

Hand it to BA-7, which prototypes in cardboard. CAD and cardboard together beat either one alone.

Who worked on this: _____ Date: _____

Build the Robot in CAD: Assembly & Fit Check

No hardware

Challenge

2-3 people

100 min

WHY THIS MATTERS

If the robot exists in CAD, you can find out whether a new mechanism fits before anyone builds it. REV publishes the Mecanum Drivetrain CAD, so you do not have to model the chassis yourself.

YOU WILL FINISH WITH

The drivetrain in an Onshape assembly with a team-designed part mated to it, and a real fit check done.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Find the Mecanum Drivetrain V2 CAD linked from REV's build guide page. Copy it into the team workspace.

20-45 min

- ☐ Create an Assembly and bring the drivetrain in. FIX it in place first, or everything you add will float around and drive you crazy.

45-75 min

- ☐ Bring in a part your team designed and mate it to the chassis exactly where it would really mount.

75-95 min

- ☐ Fit check. Does it collide with a wheel through the wheel's full rotation? Does it stick past the legal starting size? Use Onshape's measure tool on the overall dimensions.

95-100 min

- ☐ Screenshot the assembly into the notebook and tell the build crew what you found — especially if it does not fit.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ REV drivetrain CAD imported into the team workspace
- ☐ Assembly created with the chassis fixed in place
- ☐ A team-designed part mated where it would really mount
- ☐ Overall size measured in CAD and findings reported to the build crew

WHAT YOU NEED

- Onshape (CD-2)
- The Mecanum Drivetrain V2 build guide page, which links its Onshape and STEP CAD
- A part your team designed (CD-3, CD-4 or CD-7)
- Tape measure to check against the real robot

BEFORE YOU START

CD-2

THIS UNLOCKS

CD-9

IF YOU FINISH EARLY

- Measure the robot's total width and length in CAD, then measure the real robot with a tape measure. If they disagree, one of them is wrong and it is worth knowing which.

HAND OFF TO

From now on, any new mechanism gets a fit check here before the build crew cuts anything.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

A model on a screen is a drawing until it exists. Slicing and printing teaches you what CAD cannot: overhangs sag, thin walls snap, and holes always print undersized.

YOU WILL FINISH WITH

One part designed in Onshape, printed, and measured against the model to find your printer's real offset.

WHAT YOU NEED

- A model from the team library (CD-2) or your own
- 3D printer access — confirm with the coach
- Filament
- Slicer software
- Calipers

BEFORE YOU START

CD-2

THIS UNLOCKS

CD-7

IF YOU FINISH EARLY

- Reprint the same part with the hole diameter corrected by your offset, and confirm the bolt actually fits this time.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Confirm printer access and filament with the coach. Pick a SMALL part — under two hours of print time. Ambition here costs days.

15-35 min

- ☐ Export from Onshape as an STL and open it in the slicer.

35-55 min

- ☐ Set it up: layer height, infill, supports if the shape needs them, and which face goes down on the plate. Read the estimated time.
- ☐ Write every setting in the notebook. A print you cannot reproduce taught you nothing.

55-75 min

- ☐ Start the print. If it runs past the meeting, that is fine — the coach can pull it and you finish this card next meeting.

75-95 min

- ☐ Measure the printed part against the CAD model with calipers. The holes will almost certainly be undersized.
- ☐ Write down the difference. That number is your printer's real-world offset.

95-100 min

- ☐ Record the settings, the time, and the measured offset in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ STL exported and sliced with every setting written down
- ☐ Part printed
- ☐ Printed dimensions compared to the model with calipers
- ☐ The printer's offset recorded

HAND OFF TO

Write the offset number big on the whiteboard. Every part the team prints from now on should use it.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

The best use of a 3D printer on an FTC team is not decorations. It is the bracket nobody sells. Find something the robot actually needs and make it exist.

YOU WILL FINISH WITH

A printed part installed on the robot that fixes a problem the team really has.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Walk to the robot with the build crew and ask out loud: what is being held on with tape, zip ties, or hope? What bracket would we buy if it existed?
- ☐ Write down three candidates. Do not pick yet.

20-35 min

- ☐ Pick the one that is most annoying and least complicated. Measure everything it has to attach to.

35-70 min

- ☐ Model it, designing for printing this time: avoid steep overhangs, keep walls at least 3mm, put material where the load actually goes, and apply your printer offset to every hole.

70-85 min

- ❑ Slice it, check the time, and print it or queue it with the coach.

85-100 min

- ☐ Write the case for it in the notebook: what problem it solves, what it replaces, how long it took to print.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- CD-6 done, including the printer offset
- Calipers
- Access to the robot with the build crew
- Printer access

BEFORE YOU START

CD-6

THIS UNLOCKS

IF YOU FINISH EARLY

- Photograph the before and after once it is installed. This exact story — problem, design, part, result — is what judges ask for.

DONE WHEN – ALL 4 MUST BE TRUE

- ☐ Three real problems listed from looking at the actual robot
- ☐ One chosen and fully measured
- ☐ Part modeled with printability and the printer offset accounted for
- ☐ Printed or queued, with a written case for why it exists

HAND OFF TO

This is a portfolio piece. Make sure EP-5 captures it properly in the notebook.

Who worked on this: _____ Date: _____

Design the Odometry Mounting Adapter

No hardware

Challenge

2-3 people

100 min

WHY THIS MATTERS

goBILDA's 4-Bar odometry pods are built for chassis running 96-104mm drive wheels. Ours are 75mm, so our chassis sits roughly 12mm lower than the pods expect — and goBILDA's bolt pattern is not REV's. Nothing off the shelf bridges that gap. This is exactly the problem a 3D printer exists to solve.

YOU WILL FINISH WITH

A printed adapter that bolts to the REV chassis, presents goBILDA's pod pattern, and puts the pod wheel at the right height.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ MEASURE, do not assume. With calipers: the REV C-channel hole spacing and hole diameter, the pod's own mounting hole pattern, and the height from the floor to the chassis face the pod will bolt to.
- ☐ Work out the height error. The pods are designed for 96-104mm drive wheels and ours are 75mm, so roughly half that difference — about 12mm — is how much higher the pod needs to sit. Confirm it against your own measurement; do not trust the arithmetic alone.

20-45 min

- ☐ Sketch it on paper first: which face bolts to the chassis, which face the pod bolts to, and the offset between them. Two hole patterns and a height shift, nothing more.

45-80 min

- ☐ Model it in Onshape. Keep walls at least 3mm and put material along the load path — this part holds a wheel pressed against the floor for a whole match.

80-92 min

- ☐ Fit-check it in CAD against the chassis assembly from CD-5 if it exists. Check the pod's 4-bar arm still has room to swing through its full travel.

92-100 min

- ☐ Slice it. Apply your printer's hole offset from CD-6 to every bolt hole, then print it or queue it with the coach.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Every dimension measured with calipers and written down before modelling started
- ☐ Height offset calculated from a real measurement, not just the 12mm estimate
- ☐ Adapter modelled carrying both hole patterns
- ☐ Sliced with the printer offset applied, and printed or queued

WHAT YOU NEED

- Onshape (CD-2)
- Calipers
- The robot, to measure (BA-3)
- One 4-Bar odometry pod in hand to measure
- 3D printer access
- Your printer's hole offset from CD-6

BEFORE YOU START

CD-2, BA-3

THIS UNLOCKS

BA-10

IF YOU FINISH EARLY

- Print a thin test coupon of just the two hole patterns first — ten minutes of printing — and check it actually bolts up before committing to the whole adapter.

HAND OFF TO

BA-10 installs the pods using this. If it does not fit, that is normal and useful: measure what was wrong and revise.

Who worked on this: _____ Date: _____

Fit the Starter Bot to Our Chassis in CAD

No hardware

Challenge

2-3 people

100 min

WHY THIS MATTERS

REV publishes the whole Starter Bot as a public Onshape document. Their robot is not ours — their drivetrain is a two-motor channel chassis on 90mm wheels, ours is a four-motor mecanum chassis on 75mm wheels. Working out where everything lands before anybody drills is the cheapest hour of the season.

YOU WILL FINISH WITH

REV's superstructure positioned on our mecanum chassis in CAD, with every mounting change written down as a number.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Open REV's Starter Bot document and copy it into the team workspace. Never edit REV's original — work in your own copy.

20-45 min

- ☐ Measure in CAD: how far apart are the four 70mm uprights the launcher bolts to, and where do they sit along the 408mm drive rails? Write the numbers down.

45-70 min

- ☐ Bring in our chassis. Both robots use 408mm and 248mm C channel, so the upright spacing should transfer — check that it does. Where do the Ultra 90 gearboxes and the odometry pods land? Anything in the way is a problem worth finding now.

70-90 min

- ☐ Ride height. Their wheels are 90mm, ours are 75mm, so our frame sits 7.5mm closer to the floor. Every height measured from the ground has moved by that much. Decide how to make it up and put the number on the drawing.

90-100 min

- ☐ Export a marked-up view with the dimensions the build crew needs, and print it.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Team copy of REV's CAD with our chassis in it
- ☐ Upright positions dimensioned
- ☐ Gearboxes and odometry pods checked for interference
- ☐ Ride-height difference calculated and a fix chosen

WHAT YOU NEED

- Onshape account (CD-1)
- Our chassis model (CD-5)
- REV's public Starter Bot Onshape document
- Calipers

LINKS

Clickable on the Coaches Corner site

• REV Starter Bot CAD (Onshape)

cad.onshape.com/documents/9cb9eb90e973fd37d10ee178/.../a55c018c62564f3ea8812a47

BEFORE YOU START

CD-5

THIS UNLOCKS

BA-16

IF YOU FINISH EARLY

- Model the bracket that makes up the ride-height difference and print it (CD-6). A printed shim beats a stack of washers.

HAND OFF TO

The marked-up print goes straight to BA-16. That crew should not be guessing at these numbers.

Who worked on this: _____ Date: _____

Get Connected: Control Hub & Driver Hub

Board or robot

Start here

2-3 people

100 min

WHY THIS MATTERS

No programming happens until the Driver Hub talks to the Control Hub — and a programmer who only knows one way to reach the robot is stuck the day it fails. Learn both ways, and write down what you did.

YOU WILL FINISH WITH

Control Hub renamed and secured, Driver Hub connected, and the code editor reached over USB and over Wi-Fi.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Gather everything and confirm the battery is actually charged. Open the REV Hardware Client guide and keep it open the whole card.

15-35 min

- ☐ Power the Control Hub, plug it into the laptop over USB, and start the REV Hardware Client. Select the Control Hub on the Hardware tab, then Open Console. The main page shows the hub's Wi-Fi name and password.

35-50 min

- ☐ On the console's Manage page, rename the network to 22257-RC (or 22257-B-RC for the programming board, so the two hubs never share a name) and change the default password, which is literally 'password'.
- ☐ Record the new password in the notebook and tell the coach. Then connect the Driver Hub to the new network and confirm it shows the robot connected.

50-90 min

- ☐ TWO WAYS TO REACH THE CODE EDITOR — try both. Either one works.
- ☐ OVER USB, on any laptop: open the Blocks or OnBot Java tab in the Hardware Client console. No browser needed, and the internet stays up.
- ☐ OVER WI-FI: join 22257-RC, then open Chrome at 192.168.43.1:8080. With only ONE Wi-Fi adapter, joining the robot network drops the internet; with TWO, one stays on school Wi-Fi and one joins the robot.
- ☐ Find out how many Wi-Fi adapters this laptop has, and write in the notebook which ways work on it.

90-100 min

- ☐ Record the software versions the Hardware Client shows (ask the coach before any update), then write a half-page 'How to connect' guide covering both ways. Tape it inside the laptop case.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Network renamed, default password changed and shared with the coach
- ☐ Driver Hub shows the robot connected
- ☐ Editor reached over USB, and over Wi-Fi if the laptop allows it
- ☐ Connect guide written, saying which ways work on this laptop

WHAT YOU NEED

- REV Control Hub and Driver Hub
- Charged 12V battery
- USB-A to USB-C cable
- A WINDOWS laptop with the REV Hardware Client (not Mac or Chromebook)

LINKS

Clickable on the Coaches Corner site

- **REV Hardware Client guide**
docs.revrobotics.com/rev-hardware-client

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

PR-2

IF YOU FINISH EARLY

- If the laptop has only one Wi-Fi adapter, ask the coach whether the team has a USB Wi-Fi adapter. Plugged in, it becomes the second adapter, so the laptop can stay online and on the robot network at once.

HAND OFF TO

PR-2 cannot start until this works. Tell the next crew where the connect guide is.

Who worked on this: _____ Date: _____

Build the Robot Configuration File

WHY THIS MATTERS

The code cannot find a motor that is not in the configuration. Roughly half of all 'my code does not work' problems are a typo in this file.

YOU WILL FINISH WITH

A saved, activated configuration that matches the physical robot, with the port map and names recorded in the notebook.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-5 min

- ☐ With the build crew or the BA-4 port map, list every motor and servo: what it does and which hub port it is plugged into. Write it on the whiteboard.

5-10 min

- ☐ Agree on names and be boring about them: lowercase with underscores. For the mecanum drive that means `front_left_drive`, `front_right_drive`, `back_left_drive`, `back_right_drive`.
- ☐ Everyone uses these exact names from now on.

10-20 min

- ☐ On the Driver Hub: Configure Robot, create a new configuration, and add each device to its correct port with its exact name. Keep the built-in IMU on I2C bus 0 named `imu` — PR-3 needs it. Save with the team name in the filename.

20-27 min

- ☐ Activate the configuration. Spin each motor for one second and confirm the RIGHT motor moves. Fix any mismatch now.

27-30 min

- ☐ Write the port map and the naming list in the engineering notebook, and photograph the whiteboard.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Every named motor spins the correct physical motor
- ☐ Configuration saved AND activated
- ☐ Port map and naming list recorded in the notebook

WHAT YOU NEED

- Connected robot or programming board (PR-1)
- The port map from BA-4, or the build crew
- Driver Hub
- Whiteboard

BEFORE YOU START

PR-1

THIS UNLOCKS

PR-10, PR-3, PR-8

IF YOU FINISH EARLY

- Note which motors will need to be reversed in code for the robot to drive straight — PR-3 will need that.

HAND OFF TO

This card takes about 30 minutes — keep going with PR-3 (make it drive) on the robot, or PR-10 on the programming board. Give the naming list to everyone who writes code; changing a name later breaks every OpMode.

Who worked on this: _____ Date: _____

PR-3

First TeleOp: Make It Drive (Mecanum)

WHY THIS MATTERS

This is the moment robotics gets real. You will run the same mecanum drive two ways — a Blocks program from REV and a Java program from the Herobotix GitHub — so you know both tools before you write your own.

YOU WILL FINISH WITH

Two TeleOp modes on the Driver Hub, one Blocks and one Java, both driving, turning and strafing the robot.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ PROVE THE MOTORS FIRST. Prop the robot up, spin ONE motor at a time, and confirm the right corner turns the right way. Skipping this is the number one reason mecanum 'does not work'.

15-50 min

- ☐ BLOCKS: on the REV page, download the Mecanum Example Code file and upload it in the Blocks editor. Save it as TeleOp_Mecanum_Blocks.
- ☐ REV names its motors frontLeft, backLeft, frontRight, backRight. Ours are front_left_drive and so on — change every motor block's dropdown to our names.
- ☐ Run it on the floor: left stick drives and strafes, right stick turns.

50-85 min

- ☐ JAVA: in OnBot Java create a file named RobotTeleopMecanumFieldRelativeDrive and paste the Herobotix sample into it.
- ☐ Change the first line to package org.firstinspires.ftc.teamcode; and delete the @Disabled line, or it will never show on the Driver Hub.
- ☐ Check the hub orientation lines match how the Control Hub is mounted, then Build Everything. Run it: point the robot away from you and press A to zero the heading. It drives field-relative; hold the left bumper for robot-relative.

85-100 min

- ☐ Let each driver try both modes. Write in the notebook how they differ and which motors had to be reversed.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Every motor proven on its own before any drive code ran
- ☐ Blocks TeleOp drives, turns and strafes using our motor names
- ☐ Java TeleOp builds, shows on the Driver Hub, and drives field-relative
- ☐ Both saved as separate TeleOp modes, with differences noted in the notebook

WHAT YOU NEED

- Configured robot (PR-2), including the built-in IMU named imu
- Driver Hub and gamepad
- Laptop with the REV Hardware Client
- Something to prop the robot on, wheels off the ground
- Clear floor space

LINKS

Clickable on the Coaches Corner site

• REV mecanum Blocks example

docs.revrobotics.com/duo-control/hello-robot-blocks/.../programming-mecanum-refined

• Herobotix mecanum Java sample

github.com/herobotix/FtcRobotController/.../RobotTeleopMecanumFieldRelativeDrive.java

BEFORE YOU START

PR-2

THIS UNLOCKS

MF-6, PR-13, PR-14, PR-16, PR-19, PR-4, PR-5, PR-6

IF YOU FINISH EARLY

- Open the Java file and find where the field-relative math rotates the joystick input by the robot's heading. That idea is what PR-14 is all about.

HAND OFF TO

Tell the whole team — the robot drives now. Drivers can practise with whichever mode they prefer.

Who worked on this: _____ Date: _____

Telemetry: See What the Robot Is Thinking

Board or robot

Some experience

1-2 people

100 min

WHY THIS MATTERS

Telemetry prints live values to the Driver Station screen. Without it you are guessing at why something failed. With it, you can see the problem.

YOU WILL FINISH WITH

The Driver Station shows live, correct values for sticks, motor power, encoders, and time.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Read the telemetry section in the FTC Docs programming resources.

15-45 min

- ☐ Add telemetry lines for: each gamepad stick value, each motor's commanded power, and elapsed match time.

45-70 min

- ☐ Run it while someone drives and watch the screen. Do the numbers match what you expect?
- ☐ Check that the sticks read 0 when centered — if not, you have found a real problem worth fixing (see PR-5).

70-90 min

- ☐ Add encoder position readouts for the drive motors. You will need these for PR-7.

90-100 min

- ☐ Photograph the Driver Station screen into the notebook with a note about what each line means.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Working TeleOp (PR-3)
- FTC Docs telemetry section
- Driver Station

BEFORE YOU START

PR-3

THIS UNLOCKS

PR-11, PR-12, PR-14, PR-15, PR-20, PR-7

IF YOU FINISH EARLY

- Add a line that reads battery voltage and shouts a warning when it drops below a threshold you pick.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Sticks, power, encoders, and time all display and change correctly
- ☐ Centered sticks read at or near 0
- ☐ Screen photographed with notes

HAND OFF TO

Every later programming task should keep telemetry in. Tell the next crew not to delete it.

Who worked on this: _____ Date: _____

First Autonomous: Time-Based Movement

Robot + field

Some experience

2-3 people

100 min

WHY THIS MATTERS

Autonomous is just a list of instructions with timing. Start here before anything clever. This is the Wayfinder challenge in its simplest form.

YOU WILL FINISH WITH

An autonomous OpMode that drives a repeatable path with no driver input, and a measurement of how repeatable it actually is.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Draw the path on paper FIRST, as numbered steps: forward 2.0 s, stop, turn right 0.8 s, stop, forward 1.5 s, stop.

15-40 min

- ☐ Create a new Autonomous OpMode. Build ONLY the first move: drive forward, then stop. Test it three times.

40-70 min

- ☐ Add one step at a time. Test after EVERY single step. Change one number at a time — never two.

70-90 min

- ☐ Run the complete routine 5 times from the exact same starting spot. Measure how far off the ending position is each time and write down all 5.

90-100 min

- ☐ Record the final step list and the spread of the 5 runs in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Working drive code (PR-3)
- Taped field (EP-6)
- Tape measure
- Paper for the step plan

BEFORE YOU START

PR-3, EP-6

THIS UNLOCKS

PR-7

IF YOU FINISH EARLY

- Run it on a fresh battery and again on a low battery. Measure the difference. That gap is exactly why encoders exist — see PR-7.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Robot runs the full path unassisted 3 times in a row
- ☐ 5 runs measured and recorded
- ☐ Step list written in the notebook

HAND OFF TO

The measured spread tells the team whether time-based autonomous is good enough or whether PR-7 is needed.

Who worked on this: _____ Date: _____

Encoder Autonomous: Drive an Exact Distance

WHY THIS MATTERS

Time-based autonomous drifts as the battery drains, so it works in practice and fails in a match. Encoders count actual wheel rotations, so the distance stays honest.

YOU WILL FINISH WITH

A routine that drives a commanded number of inches, landing within about an inch, five times running.

WHAT YOU NEED

- Working time-based auto (PR-6)
- Encoder telemetry (PR-4)
- Tape measure
- Motor spec sheet
- Calculator
- gm0 + FTC Docs

BEFORE YOU START

PR-6, PR-4

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Add an encoder-based turn, or use the Control Hub's built-in IMU to turn an exact number of degrees.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Read the encoder section in FTC Docs and gm0.
- ☐ Find your motor's counts per revolution (check the REV spec sheet for your exact motor) and measure your wheel diameter.

20-40 min

- ☐ Calculate counts per inch: (counts per motor revolution multiplied by gear ratio) divided by (wheel circumference in inches).
- ☐ Show the whole calculation on the whiteboard so someone else can check it.

40-75 min

- ☐ Write a drive-to-distance routine using RUN_TO_POSITION, or by reading encoder counts and stopping at the target.

75-95 min

- ☐ Command exactly 24 inches. Measure the real distance with a tape measure. Repeat 5 times, recording every result.
- ☐ Adjust your counts-per-inch constant based on what you measured, then re-test.

95-100 min

- ☐ Write the final counts-per-inch number large on the whiteboard and in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Commanded 24 in lands within about 1 in, five times in a row
- ☐ The counts-per-inch math is shown and checked
- ☐ Final constant recorded

HAND OFF TO

This constant is a team asset. Anyone writing autonomous should be able to find it in 10 seconds.

Who worked on this: _____ Date: _____

Intake Control: Map a Mechanism to a Button

Robot

Some experience

2 people

100 min

WHY THIS MATTERS

A driver has to run the intake without looking down at the controller. Getting the button mapping and the power level right is what makes the difference between grabbing Pollen and throwing it.

YOU WILL FINISH WITH

Intake runs in and out on gamepad buttons at a tested power level, with a control map taped to the Driver Station.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Confirm the intake motor is in the configuration and spins the correct direction. Fix it in the config, not in code, if it is backwards.

15-45 min

- ☐ Map intake-in and intake-out to two buttons. Start at LOW power — you can always turn it up.

45-70 min

- ☐ Test with real Pollen. Find the power level that grabs reliably without jamming or launching it across the room. Write that number down.

70-90 min

- ☐ Add telemetry showing the intake state and power. If the motor stalls when jammed, add a guard that cuts power.

90-100 min

- ☐ Draw a gamepad control map — a picture of the controller with every used button labeled. Tape it to the Driver Station.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Built intake (BA-5)
- Configured intake motor (PR-2)
- Real Pollen to test with
- Paper for the control map

BEFORE YOU START

PR-2, BA-5

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Make it hold-to-intake, release-to-hold, so the driver never has to think about turning it off.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Intake picks up Pollen reliably on a button press
- ☐ Working power level recorded in the notebook
- ☐ Control map drawn and taped to the Driver Station

HAND OFF TO

Show the drivers the control map and let them try it before you call it done.

Who worked on this: _____ Date: _____

Practice Without the Robot: FTC SIM

Start here

100 min

There is one robot and more than one person who wants to program. The browser simulators let everyone write and test code at the same time, with no waiting and nothing to break.

Everyone in the crew has driven and programmed a virtual robot through a course.

0-15 min

- ☐ Open ftcsim.org (and vrobotsim.online as a second option). Get everyone onto a machine.

15-40 min

- ☐ Work through the intro tutorial. Drive the virtual robot manually first so you know what good driving looks like.

40-75 min

- ☐ Program a simple autonomous in the simulator: forward, turn, forward, stop. Same discipline as the real robot — one change at a time.

75-95 min

- ☐ Race each other. Whose simulated autonomous gets furthest through the course?

95-100 min

- ☐ Write down one thing the simulator taught you that you want to try on the real robot.

NOTES, SKETCHES, WHAT SURPRISED YOU

- One laptop or Chromebook per person
- ftcsim.org and/or vrobotsim.online
- No robot required

Nothing — you can start this today

1

- Rebuild your real Wayfinder path in the simulator and compare how it behaves against the real thing.

- Everyone has a working simulated autonomous
- One transferable idea written down for the real robot

This is the best task to run in parallel when the robot is tied up in a build task.

Who worked on this: _____ Date: _____

PR-10

Touch Sensor: The First Decision

WHY THIS MATTERS

Every interesting program makes a decision. A touch sensor is the simplest one there is — pressed or not pressed. Once you can write 'if this, then that', you can write an arm that stops at the top instead of tearing itself apart.

YOU WILL FINISH WITH

A program where pressing the touch sensor changes what the hardware does, with the sensor state live on telemetry.

WHAT YOU NEED

- Assembled programming board (BA-9)
- Touch sensor in the configuration (PR-2)
- FTC Docs digital sensor pages

BEFORE YOU START

PR-2

THIS UNLOCKS

PR-13

IF YOU FINISH EARLY

- Add a counter that displays how many times the sensor has been pressed. Then work out why one push sometimes counts as several, and fix it. The word to look up is 'debounce'.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Confirm the touch sensor is in the configuration under a clear name like touch_sensor.
- ☐ Read the FTC Docs page on digital sensors.

15-40 min

- ☐ Write a program that does NOTHING but print the sensor state to telemetry. Press it, release it, watch the value flip.
- ☐ Do not move on until the number on screen matches exactly what your finger is doing.

40-70 min

- ☐ Add the decision: if the sensor is pressed, run the motor; if it is not, stop the motor. Test it.
- ☐ Now invert it, so the motor runs only when NOT pressed. Doing both is how you learn which is which.

70-90 min

- ☐ Make it useful. Use the touch sensor as a limit: the motor may run one direction freely, but stops the instant the sensor is pressed.
- ☐ This is exactly how a real arm or lift protects itself from driving into its own frame.

90-100 min

- ☐ Write in the notebook, in your own words, what if/else means, and name one place on a competition robot where you would put a limit switch.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Telemetry shows the sensor state changing correctly
- ☐ Motor behavior changes when the sensor is pressed
- ☐ Limit behavior stops the motor in one direction only
- ☐ Explanation written in the notebook

HAND OFF TO

PR-13 moves this logic onto the real robot.

Who worked on this: _____ Date: _____

Color Sensor: Teach It to See Pollen

WHY THIS MATTERS

This season's scoring element is Pollen. A robot that can tell whether it is actually holding one can make decisions the driver would otherwise have to make under pressure, while looking at the wrong thing.

YOU WILL FINISH WITH

A program that reliably reports whether a Pollen is in front of the sensor, using thresholds you measured rather than guessed.

WHAT YOU NEED

- Assembled programming board (BA-9)
- Color sensor in the configuration
- Real Pollen
- 3-4 other objects to test against
- FTC Docs color sensor pages
- Paper for your data table

BEFORE YOU START

PR-4

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Lighting changes everything for a color sensor. Test yours next to a window and again under only the room lights, and see how far your numbers move. That is why teams shroud their sensors.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Confirm the color sensor is in the configuration. Read the FTC Docs page on the color sensor.

15-45 min

- ☐ Print the raw values to telemetry: red, green, blue, and the proximity reading.
- ☐ Hold a Pollen in front of it. Then your hand, the table, a sheet of white paper. WRITE DOWN the numbers for each one.
- ☐ You are collecting data here, not guessing. The table you build is the whole point of this hour.

45-70 min

- ☐ Look at your table. Pick a threshold that separates 'a Pollen is there' from 'nothing is there'.
- ☐ Write the rule as an English sentence BEFORE you write it as code.

70-90 min

- ☐ Code the rule. Test it 10 times with a Pollen and 10 times without, counting how many times it was right.
- ☐ Under 18 correct out of 20 means your threshold is wrong — go back to your numbers, do not just nudge the code.

90-100 min

- ☐ Record the final thresholds and the 20-trial result in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ A written table of measured values for at least 4 different objects
- ☐ A threshold rule written in plain English before it was written in code
- ☐ 18 or more correct out of 20 trials
- ☐ Thresholds and results recorded

HAND OFF TO

Note that these thresholds are only valid for THIS sensor at THIS distance. Say so next to the numbers.

Who worked on this: _____ Date: _____

Potentiometer & Servo: Analog In, Position Out

WHY THIS MATTERS

A motor spins. A servo goes to a position and holds it. Confusing the two is the most common beginner mistake in FTC, and wiring a knob to a servo makes the difference obvious in about ten seconds.

YOU WILL FINISH WITH

Turning the potentiometer knob sweeps the servo to a matching position, with both values on screen.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Confirm both are in the configuration. Read the FTC Docs pages on analog input and on servos.

15-40 min

- ☐ Print the potentiometer's raw value to telemetry. Turn the knob slowly from one stop to the other and write down the value at each end.
- ☐ That pair of numbers is your input range. Everything else today depends on it.

40-60 min

- ☐ Separately, command the servo to fixed positions (0.0, 0.5, 1.0) on button presses. Watch it move there and HOLD.
- ☐ Compare that to the motor, which only spins. Say the difference out loud to your partner.

60-85 min

- ☐ Connect them: map the potentiometer's measured range onto the servo's 0 to 1 range, so turning the knob sweeps the servo. Show both the raw and the mapped value on telemetry.

85-100 min

- ☐ Write in the notebook, in your own words, the difference between a motor and a servo and where each one belongs on a competition robot.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Assembled programming board (BA-9)
- Potentiometer and servo in the configuration
- FTC Docs analog input and servo pages

BEFORE YOU START

PR-4

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Add limits so the servo can never be commanded past a safe range even if the knob turns further. Real mechanisms break when code forgets this.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Knob sweeps the servo smoothly across its full travel
- ☐ Both raw and mapped values shown on telemetry
- ☐ Motor versus servo explained in writing

HAND OFF TO

This mapping technique works for any analog sensor. PR-13 moves it to the robot.

Who worked on this: _____ Date: _____

Bench to Robot: Move Your Code Over

WHY THIS MATTERS

Code that works on the bench does not automatically work on the robot — different ports, different names, different mechanical reality. Making that jump is a real engineering skill, and it is where the two halves of your team finally meet.

YOU WILL FINISH WITH

One behavior developed on the programming board, running on the actual robot.

WHAT YOU NEED

- A working bench program (PR-10, PR-11 or PR-12)
- A robot with working TeleOp (PR-3) — either 22257's robot or the Herobotix class robot; both run a REV Control Hub
- Both port maps side by side
- A sensor you can spare to mount on the robot
- Hex drivers

BEFORE YOU START

PR-3, PR-10

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Teach the change back to whoever wrote the original bench version. Explaining it is how you find out whether you actually understood it.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Pick ONE behavior to move over — not all of them.
- ☐ Put the bench port map and the robot port map next to each other and find every name and port that differs.
- ☐ If you are using the class robot, write down WHICH robot you configured — a config saved to one hub does not exist on the other.

15-40 min

- ☐ Mount the sensor on the robot somewhere it can actually do its job, and wire it.
- ☐ Add it to the ROBOT's configuration using the same name you used on the bench if you can. That alone removes a whole category of bugs.

40-70 min

- ☐ Copy the behavior into the robot's TeleOp. Expect it not to work the first time — that is normal, not failure.
- ☐ Use telemetry to find out whether the sensor is reading at all BEFORE you start blaming the logic.

70-90 min

- ☐ Test it on the robot in a realistic situation, not just sitting on the bench. Sensor readings will be different out there — retune the thresholds against the real thing.

90-100 min

- ☐ Write down what had to change between bench and robot, and why.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ The behavior runs on the actual robot
- ☐ Telemetry confirms the sensor reads correctly where it is mounted
- ☐ A written list of what changed from bench to robot

HAND OFF TO

This is the pattern for the rest of the season: prototype on the bench, prove it on the robot.

Who worked on this: _____ Date: _____

Field-Centric Drive with the IMU

WHY THIS MATTERS

In normal driving, 'forward' means the robot's forward — so the moment the robot faces the driver, left and right swap and everyone gets lost. Field-centric drive uses the Control Hub's built-in IMU so 'forward' always means away from the driver. On a mecanum robot it is the single biggest driving upgrade available.

YOU WILL FINISH WITH

A drive mode where pushing the stick away from you moves the robot away from you, whichever way it is facing.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Read the FTC Docs pages on the Control Hub's built-in IMU.
- ☐ Print the robot's heading to telemetry and rotate the robot by hand while watching it. Confirm it reads about 0 when the robot faces your chosen 'forward'.

20-45 min

- ☐ Understand the idea before you code it: you rotate the driver's stick input by the **NEGATIVE** of the robot's current heading, then feed that result into the same mecanum mixing you wrote in PR-3.
- ☐ Draw it on paper with the robot turned 90 degrees before you touch the keyboard.

45-80 min

- ☐ Implement it. Add a button that resets the heading to zero — the driver needs it at the start of every match and you will need it constantly while testing.

80-95 min

- ☐ Test properly: point the robot in four different directions and each time push the stick straight away from you. It should move away from you every single time.

95-100 min

- ☐ Record what changed and let two different drivers try both modes.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Working mecanum TeleOp (PR-3)
- Telemetry (PR-4)
- Taped field (EP-6)
- FTC Docs IMU pages
- Two drivers to test with

BEFORE YOU START

PR-3, PR-4

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Put robot-centric and field-centric on a toggle button and let the drivers pick which they want for Fast Lane. Record which they chose and why.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Heading displays on telemetry and behaves correctly
- ☐ Reset-heading button works
- ☐ Robot moves away from the driver from all four starting orientations

HAND OFF TO

This is the mode most competitive teams drive with. Note which drivers prefer which mode.

Who worked on this: _____ Date: _____

Read the Pinpoint: Configure & Prove It

Robot

Some experience

2-3 people

100 min

WHY THIS MATTERS

A sensor the code cannot read is decoration. This card gets the Pinpoint into the configuration, its numbers onto the screen, and then proves those numbers are actually true before anything gets built on top of them.

YOU WILL FINISH WITH

The Pinpoint configured, reporting position on telemetry, and validated against a tape measure.

WHAT YOU NEED

- Robot with the Pinpoint installed (BA-10)
- The pod offsets from BA-10
- goBILDA's Java driver and sample OpMode — github.com/goBILDA-Official/FtcRobotController-Add-Pinpoint
- Tape measure
- Telemetry (PR-4)

BEFORE YOU START

BA-10, PR-4

THIS UNLOCKS

PR-16

IF YOU FINISH EARLY

- Rotate the robot one full turn counterclockwise. The heading should read very nearly one full rotation positive. If it does not, your pod offsets are wrong — that test catches it faster than anything else.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Add the Pinpoint to the configuration as an I2C device on the port BA-10 used, and add goBILDA's driver and sample OpMode to the project.
- ☐ IMPORTANT: the Pinpoint forgets its settings on every power cycle, so your code must send the configuration in init on every single run.

20-40 min

- ☐ Set the encoder resolution. With goBILDA 4-Bar pods you just tell the driver which pod you have and the ticks-per-mm maths is already done for you.
- ☐ Set the two pod offsets from BA-10, in millimetres, with the signs the user guide describes.

40-80 min

- ☐ Put X, Y and heading on telemetry. Call `resetPosAndIMU()` with the robot completely still, then watch the numbers untouched — any drift means the IMU zeroed while something was moving.
- ☐ DIRECTION TEST. Push the robot straight forward: X must INCREASE. Push it LEFT: Y must INCREASE.
- ☐ If either runs backwards, reverse that encoder's direction in the configuration — never by flipping a sign in your own maths, where nobody will find it.

80-95 min

- ☐ PROVE IT. Tape-measure a known distance, push the robot along it, and compare against what the Pinpoint reports. Then drive a loop back to the exact start — a healthy setup lands within about 10mm, and much more means something is wrong.

95-100 min

- ☐ Record the offsets, the encoder directions and your loop-closure error in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Pinpoint in the configuration and reconfigured at init on every run
- ☐ Numbers hold still when the robot is still; X increases forward and Y increases left
- ☐ Reported distance matches a tape measure
- ☐ A drive-and-return loop closes within about 10mm

HAND OFF TO

Pedro Pathing (PR-16) is built entirely on this. Do not move on with a bad loop-closure number; you will only find it again later, slower.

Who worked on this: _____ Date: _____

Pedro Pathing: Install & Tune the Localizer

Robot + field

Challenge

2-3 people

100 min

WHY THIS MATTERS

Pedro Pathing turns 'drive forward two seconds' into 'follow this curve to that point on the field'. It is how competitive teams write autonomous. But it can only steer toward a position it trusts, so the localizer gets tuned first and everything else waits its turn.

YOU WILL FINISH WITH

Pedro Pathing installed with the Pinpoint as its localizer, passing Pedro's own localization test.

WHAT YOU NEED

- A robot that passes PR-15
- Pedro Pathing docs at pedropathing.com
- Laptop on the robot's network
- Taped field (EP-6)
- Clear floor space to push the robot around

BEFORE YOU START

PR-15, PR-3

THIS UNLOCKS

PR-17

IF YOU FINISH EARLY

- Deliberately set one pod direction wrong and watch the localization test fall apart, then put it back. Knowing what broken looks like will save you an hour at a competition.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Install Pedro Pathing into the project following the docs. Add the PinpointTuner to your Tuning class with the @Tuner annotation.

20-35 min

- ☐ Connect to the Robot Controller's tuning page at <http://192.168.43.1:10158> and open the Pinpoint procedure.
- ☐ Enter the Pinpoint's hardware map name and choose the goBILDA pod type.

35-60 min

- ☐ Direction test, Pedro's version: push the robot forward to set the X pod direction, then push it left to set the Y pod direction. Same physics as PR-15, different tool.

60-80 min

- ☐ Offset calibration: rotate the robot 180 degrees counterclockwise and let the tuner compute the pod offsets itself.
- ☐ COMPARE what it computes against what you measured by hand in BA-10. They should be close. If they are not, stop and find out which one is lying before you continue.

80-95 min

- ☐ Copy the generated PinpointConfig values into Constants.java. Run Pedro's localization test and drive the robot around the field watching the reported position.

95-100 min

- ☐ Record the final values, and a note on how closely the tuner's offsets matched your hand measurements.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Pedro Pathing installed and the project builds
- ☐ PinpointConfig generated and pasted into Constants.java
- ☐ Tuner-computed offsets agree with the ones measured in BA-10
- ☐ Pedro's localization test tracks the robot correctly around the field

HAND OFF TO

PR-17 tunes the follower. Do not start it until localization is genuinely solid — every tuner after this assumes the position is right.

Who worked on this: _____ Date: _____

Pedro Pathing: Tune the Follower

WHY THIS MATTERS

Localization tells the robot where it is. The follower decides what to do about it. These tuners run in a fixed order because each one builds on the last, and skipping ahead gives you a robot that oscillates its way down the field while everyone argues about why.

YOU WILL FINISH WITH

A robot that follows a Pedro path smoothly and stops where it was told to stop.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Weigh the robot and put its real mass into Constants. Confirm the drivetrain settings match your mecanum setup.
- ☐ Read the whole tuning order before touching anything: mass, then the automatic tuners, then the three PIDFs, then centripetal. In that order, every time.

15-45 min

- ☐ Run the automatic tuners in order: Forward Velocity, Lateral Velocity, Forward Zero Power Acceleration, then Lateral Zero Power Acceleration.
- ☐ Each wants a clear run of floor. Write every number down as it comes out — they are hard to recover later.

45-70 min

- ☐ Translational PIDF. The goal is straight-line movement with no sideways drift. Change ONE gain at a time and watch what it actually does before touching another.

70-90 min

- ☐ Heading PIDF, then Drive PIDF. More drive PIDF means faster along the path but more overshoot at the end; less means slower and tighter. Decide which trade you want and tune toward it deliberately.

90-100 min

- ☐ Record every tuned value in the notebook with today's date. These numbers are this robot at this weight — change the robot and they change too.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Mass set from an actual weighing, not a guess
- ☐ All four automatic tuners run and their values recorded
- ☐ Translational, heading and drive PIDFs tuned in that order
- ☐ The robot follows a test path and stops on target

WHAT YOU NEED

- PR-16 complete
- The biggest clear floor space you have
- A scale to weigh the robot
- Tape measure
- Pedro Pathing tuning docs
- Patience — this card may well run into a second meeting

BEFORE YOU START

PR-16, EP-6

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Run the centripetal tuning and then drive a curved path. A robot that corners properly is dramatically faster than one that drives straight lines and stops to turn.

HAND OFF TO

Retune after any significant weight change — a new mechanism changes all of it. Date the values so the next crew knows how stale they are.

Who worked on this: _____ Date: _____

Starter Bot Configuration: Six Motors, Two Hubs

Robot

Some experience

2-3 people

100 min

WHY THIS MATTERS

REV's sample code expects exact device names. Our robot has six motors instead of four, because mecanum uses all four Control Hub ports and the intake and flywheel have to live on the Expansion Hub. Get the names and ports right and the sample runs; get them wrong and nothing works and nothing tells you why.

YOU WILL FINISH WITH

A saved, activated configuration with six motors and two servos, each one proved on its own.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Write the plan on the whiteboard before touching the software. Control Hub ports 0-3: the four mecanum drive motors, named exactly as in PR-2. Expansion Hub port 0: intakeMotor (Core Hex). Expansion Hub port 1: flywheel (UltraPlanetary HD Hex). Servos: crServo on 0, pollenServo on 1.

15-35 min

- ☐ In the configuration screen, scan for the Expansion Hub. If it does not appear, the problem is the communication cable or the hub address — not the code.

35-65 min

- ☐ Enter every device with the exact name and type. Spelling and capitals matter: flywheel is not Flywheel.

65-90 min

- ☐ Prove each device alone, with the hub's motor test or a short OpMode. Wrong direction is fine for now; wrong device is not.

90-100 min

- ☐ Save, activate, photograph both configuration screens, and write the port map in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Both hubs visible in the configuration
- ☐ Six motors and two servos named exactly as the sample expects
- ☐ Every device moves when its own name is called
- ☐ Config saved, activated and photographed

WHAT YOU NEED

- Robot with Expansion Hub wired (BA-17)
- Driver Hub or REV Hardware Client
- Laptop
- REV's configuration table

LINKS

Clickable on the Coaches Corner site

- **REV Starter Bot config, wiring and gamepad map**
docs.revrobotics.com/ftc-kickoff-concepts/home/programming-teleop-and-auto

BEFORE YOU START

BA-17

THIS UNLOCKS

PR-19

IF YOU FINISH EARLY

- Export the configuration file off the hub onto the team laptop. When a hub gets wiped the day before an event, you will have it back in two minutes.

HAND OFF TO

Give the port map to PR-19 and PR-21.

Who worked on this: _____ Date: _____

Run the Starter Bot TeleOp on Mecanum

Robot + field

Challenge

2-3 people

100 min

WHY THIS MATTERS

REV ships working TeleOp for the Starter Bot in both Blocks and OnBot Java — but it drives a two-motor arcade drivetrain. Everything above the wheels works as-is; the drive half is ours to replace with the mecanum code we already wrote in PR-3. That swap is the most valuable programming job of the fall.

YOU WILL FINISH WITH

One TeleOp that drives mecanum and runs the intake, flywheel, feeder and pollen pusher.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Download REV's TeleOp — Blocks or OnBot Java, whichever the crew already knows — and load it unchanged. Read it together and find the two drive lines.

15-45 min

- ☐ Replace the arcade drive with the mecanum mixing from PR-3. Four motors instead of two. Change nothing else.

45-70 min

- ☐ Drive it with the mechanisms switched off. Prove strafing still works and that nothing is reversed.

70-90 min

- ☐ Test the mechanisms one at a time against REV's gamepad map: right bumper intakes, right trigger reverses, left bumper runs the flywheel sequence, A and B move the pollen pusher.

90-100 min

- ☐ Score one Pollen into the Hive. Write down every button that did something you did not expect.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Robot strafes and turns under driver control
- ☐ Intake, flywheel, feeder and pusher all respond
- ☐ At least one Pollen scored
- ☐ Gamepad map written in the notebook

WHAT YOU NEED

- Robot (BA-16, BA-17)
- Configuration from PR-18
- Practice field and Pollen
- Gamepad
- The mecanum code from PR-3

LINKS

Clickable on the Coaches Corner site

- **REV Starter Bot TeleOp and Auto**

docs.revrobotics.com/ftc-kickoff-concepts/home/programming-teleop-and-auto

BEFORE YOU START

PR-18, PR-3

THIS UNLOCKS

EP-9, PR-20, PR-21

IF YOU FINISH EARLY

- Try REV's split-arcade idea on the mecanum version — one stick to move, one to turn. Ask the drivers which they prefer before committing.

HAND OFF TO

This OpMode is the drive team's practice tool. Hand it to EP-9 and do not change it without telling them.

Who worked on this: _____ Date: _____

Flywheel Tuning: Make the Shot Repeatable

Robot + field

Challenge

2-3 people

100 min

WHY THIS MATTERS

A launcher that scores nine times out of ten wins matches. A launcher run on raw power scores when the battery is fresh and misses when it is not. Velocity control is what makes the tenth shot land where the first one did.

YOU WILL FINISH WITH

A target flywheel velocity that scores from a marked spot, and proof it still works on a half-drained battery.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Switch the flywheel from power to velocity control. The launcher motor is an HD Hex on a 1:1 gearbox — work out its maximum ticks per second and put that number on the board.

15-40 min

- ☐ Put telemetry on the screen: target velocity, actual velocity, and whether it is up to speed. You cannot tune what you cannot see.

40-70 min

- ☐ Tape a shooting spot. Fire five Pollen at one velocity and count the makes, then change velocity and repeat. Build a table on paper: distance, velocity, makes out of five.

70-85 min

- ☐ Add a second spot at a different distance and fill in that row.

85-95 min

- ☐ Swap in a battery that has been used a while and re-shoot the first spot. If the result changes, the code is feeding before the flywheel is up to speed.

95-100 min

- ☐ Write the winning numbers into the code as named constants, and into the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Velocity control working with telemetry on screen
- ☐ A table of distance vs velocity vs makes out of five
- ☐ Same result on a fresh and a used battery
- ☐ Constants named in the code, values in the notebook

WHAT YOU NEED

- Robot and practice field
- 10+ Pollen
- Tape to mark shooting spots
- Telemetry from PR-4
- One fresh battery and one used one

BEFORE YOU START

PR-19, PR-4

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Make the code refuse to feed until the flywheel is within a few percent of target. That one rule fixes most 'it worked yesterday' problems.

HAND OFF TO

Give the distance table to the drive team (EP-9) and tape the shooting spots onto the practice field.

Who worked on this: _____ Date: _____

Starter Bot Autonomous: Score the Pre-Loads

Robot + field

Challenge

2-3 people

100 min

WHY THIS MATTERS

Thirty seconds of autonomous are worth real points and most rookie teams score none of them. REV ships an Auto that launches pre-loaded Pollen and is meant to be customised for either starting side. Getting it running is the cheapest scoring on this robot.

YOU WILL FINISH WITH

An autonomous that starts from either side, scores the pre-loads, and finishes without breaking a rule.

WHAT YOU NEED

- Robot and practice field
- Legal starting positions marked
- Pollen
- REV's Auto sample
- Timer

BEFORE YOU START

PR-19

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Park at the end if there are seconds to spare. Points you already have are worth more than points you might get.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Load REV's Autonomous sample and read it together. Find where the starting side is chosen and where the launch velocity comes from.

15-35 min

- ☐ Mark the legal starting positions on the field with tape, measured from the Competition Manual. Guessing here costs the whole auto.

35-65 min

- ☐ Run it from one side and time it. If it runs long, it is usually flywheel spin-up rather than driving.

65-85 min

- ☐ Make the other side work — a flag or a second OpMode, whichever the crew can explain to a judge.

85-95 min

- ☐ Run each side five times without touching anything in between. Count how many score.

95-100 min

- ☐ Write both routines and their scores into the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Auto runs from both starting sides
- ☐ Scores the pre-loads at least three times out of five
- ☐ Finishes inside the autonomous period
- ☐ Robot legal in its starting position

HAND OFF TO

Tell the drive team which side the auto is better on. That is a real thing to offer an alliance partner (EP-8).

Who worked on this: _____ Date: _____

WHY THIS MATTERS

You cannot build with parts you cannot find. This is the highest-value hour a brand-new team ever spends, and it pays back every single meeting afterward.

YOU WILL FINISH WITH

Everything counted, labeled, and in a known place, with a written inventory and a photo of what clean looks like.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Clear a big table. Open the REV kit and pull out any existing team bins.

15-50 min

- ☐ SORT FIRST, count later. Group by type: structure and channel, motors, wheels, electronics, fasteners, tools, miscellaneous.

50-75 min

- ☐ Now count. Record into the inventory sheet: part name, quantity, and where it lives.

75-90 min

- ☐ Label every bin and drawer. Then take a photo of the shelf looking exactly the way it should look.

90-100 min

- ☐ Print the photo and post it above the shelf. That photo is now the team's cleanup standard — the last 10 minutes of every meeting is matching it.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- The REV kit and every existing parts bin
- Bins, drawers, or containers
- Label tape or a label maker
- Laptop or paper for the inventory sheet

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

BA-11, BA-2

IF YOU FINISH EARLY

- Flag anything you are low on and add it to the team's parts wish list for the coach.

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ Inventory sheet filled in with quantities and locations
- ☐ Every bin labeled
- ☐ Reference photo printed and posted above the shelf

HAND OFF TO

Show the whole team where things live.
Point at the reference photo during the first cleanup.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

The drivetrain is the foundation — every other mechanism bolts to it. On a mecanum robot a frame that is not square means four wheels that fight each other instead of working together.

YOU WILL FINISH WITH

A square C-channel chassis built to the REV Mecanum Drivetrain Kit v2 build guide.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Open the Mecanum Drivetrain V2 build guide and read the WHOLE frame section before anyone picks up a tool.
- ☐ The guide splits the build into sections so people can work in parallel — decide now who takes which section.

15-30 min

- ☐ Lay out the C-channels and the brackets and fasteners the frame section calls for.
- ☐ SORT THE CHANNELS BY LENGTH FIRST. The kit has two lengths (408mm and 248mm) and mixing them up costs you an hour.

30-75 min

- ☐ Assemble the frame following the guide. Start every fastener by hand, then snug it — do NOT fully tighten until the frame is square.

75-90 min

- ☐ Check square: measure both diagonals of the frame. Equal diagonals means square. Adjust until they match, THEN tighten everything down.

90-100 min

- ☐ Photograph the finished frame into the notebook with the date and who built it.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- REV Mecanum Drivetrain Kit v2 (REV-45-2470)
- The Mecanum Drivetrain V2 build guide — link on the resource shelf
- Hex drivers (included in the kit)
- Tape measure
- Safety glasses

BEFORE YOU START

BA-1, EP-7

THIS UNLOCKS

BA-3

IF YOU FINISH EARLY

- Start a torque-check habit: at the start of every meeting, someone re-checks every fastener on the robot.

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ Diagonals measured within about 1/8 in of each other
- ☐ All fasteners tightened after squaring, not before
- ☐ Channel lengths match the guide, no unexplained leftover parts

HAND OFF TO

Hand off to BA-3 for gearboxes, motors and wheels.

Who worked on this: _____ Date: _____

Mecanum Drive: Gearboxes, Motors & Wheels

Robot

Some experience

3-4 people

100 min

WHY THIS MATTERS

This is where a frame becomes a robot that can move in any direction. It is also where mecanum goes wrong: the four wheels are NOT interchangeable, and one wheel in the wrong corner turns strafing into spinning.

YOU WILL FINISH WITH

Four motor and gearbox assemblies mounted with the mecanum wheels in the correct pattern, every wheel spinning freely.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Read the gearbox and wheel sections of the build guide.
- ☐ Each corner is an HD Hex motor with an UltraPlanetary gearbox plus an Ultra 90 to turn the output 90 degrees. Build ONE complete assembly together as a group first, so everybody has seen how it goes together, then split up.

20-60 min

- ☐ Build the remaining three assemblies and mount all four to the frame per the guide. Keep the left and right sides correctly mirrored.

60-80 min

- ☐ MECANUM WHEEL CHECK — this is the mistake every new team makes. The four wheels are handed and they are not interchangeable.
- ☐ Mount them, then look straight down at the robot from above. The top rollers should form an X across the chassis.
- ☐ If you see rollers running parallel, or a diamond instead of an X, two wheels are swapped. Check against the build guide photos or the Onshape CAD before you accept it.

80-95 min

- ☐ Spin every wheel by hand — anything that binds or rubs gets fixed NOW, not later.
- ☐ Set the robot on a flat floor and confirm all four wheels touch at the same time.

95-100 min

- ☐ Write down which motor is in which corner. The programming crew needs this for PR-2.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Four motor and gearbox assemblies mounted
- ☐ Rollers form an X viewed from above, verified against the guide
- ☐ Every wheel spins freely and the robot sits flat
- ☐ Corner positions written down for the programmers

WHAT YOU NEED

- Squared frame (BA-2)
- Mecanum Drivetrain V2 build guide
- 4x UltraPlanetary gearbox kits with HD Hex motors
- 4x Ultra 90 degree gearboxes
- The 75mm mecanum wheel set
- Hex drivers
- A flat floor

BEFORE YOU START

BA-2

THIS UNLOCKS

BA-4, BA-5, CD-8

IF YOU FINISH EARLY

- Push the robot sideways by hand across the floor. If the wheels are right it slides sideways smoothly. If it fights you or twists, re-check the X pattern before you blame the code later.

HAND OFF TO

Give the corner positions to whoever does PR-2. If the X pattern is wrong, strafing will never work no matter how good the code is.

Who worked on this: _____ Date: _____

Electronics: Mount & Wire the Control Hub

Robot

Some experience

2-3 people

100 min

WHY THIS MATTERS

The Mecanum Drivetrain Kit includes no control system, no battery and no wiring — all of that is on you. Bad wiring is the number one reason a robot dies mid-match, and on a four-motor mecanum drive a single mislabeled cable turns strafing into spinning.

YOU WILL FINISH WITH

Control Hub, battery and switch mounted and wired, all four drive motors labeled at both ends and tied down.

WHAT YOU NEED

- Robot with drivetrain (BA-3)
- A REV Control Hub — NOT included in the kit; confirm with the coach that one is free before you start
- 12V battery and main power switch
- Motor cables and power cabling
- Zip ties
- Label tape
- Coach or mentor

BEFORE YOU START

BA-3, EP-7

THIS UNLOCKS

BA-10, BA-17

IF YOU FINISH EARLY

- Make a laminated port map card that lives on the robot cart, so a mid-competition repair does not need a laptop.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Confirm with the coach that a Control Hub is actually available for this robot. The kit does not include one and the team's hubs are already spoken for — do not start until this is settled.
- ☐ Read the wiring guidance in the build guide. ASK THE COACH before cutting, crimping or modifying any cable.

20-55 min

- ☐ Mount the Control Hub where you can reach the ports AND see the status lights. Mount the battery so it physically cannot fall out or slide, and put the switch somewhere a referee can reach it.

55-80 min

- ☐ Route all four motor cables to the hub. Label BOTH ENDS of every cable: LF, RF, LB, RB.
- ☐ On a mecanum drive the corner labels ARE the robot's behavior — get them wrong and the strafe math in PR-3 will be chasing a wiring bug.

80-95 min

- ☐ Tie down every cable so nothing can reach a wheel or a moving part. Tug-test each connector gently to confirm it is seated.

95-100 min

- ☐ Photograph the wiring and write the port map into the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Control Hub, battery and switch mounted, battery cannot fall out
- ☐ All four motor cables labeled by corner at both ends
- ☐ Nothing loose, hub status lights visible, port map recorded

HAND OFF TO

The port map goes straight to PR-2. Any wiring change means the configuration changes too.

Who worked on this: _____ Date: _____

Design & Build an Intake

WHY THIS MATTERS

The Mecanum Drivetrain Kit is a drive base and nothing else — it ships with no intake and there are no instructions to follow. Whatever picks up Pollen on this robot, your team designs and builds. That is real FTC, and it is the part judges ask about.

YOU WILL FINISH WITH

A mounted intake prototype that picks up a Pollen, with a measured success rate.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Review the concept chosen in EP-4 and gather parts. Check the Competition Manual sizing rule — whatever you add has to keep the robot legal at the start of a match.

15-45 min

- ☐ Build the MOUNTING first — how does this attach to the C-channel? Get that solid before you worry about the grabbing part. A great intake that falls off is worth nothing.

45-75 min

- ☐ Build the grabbing mechanism. Rough is fine at this stage; cardboard and zip ties are legitimate. Test it by hand with a Pollen as you go.

75-95 min

- ☐ Power it if you have a spare motor or servo. Do 10 pickup attempts and count how many succeed. Write the number down.

95-100 min

- ☐ Photograph it and record the mounting height and the success count — you will change the height later and need to know where you started.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Robot with drivetrain (BA-3)
- The concept chosen in EP-4
- Parts from the team's stock
- Cardboard and zip ties for mock-ups
- Real Pollen (10+)
- Hex drivers

BEFORE YOU START

BA-3, EP-4

THIS UNLOCKS

BA-7, PR-8

IF YOU FINISH EARLY

- Test a second mounting height, 10 more attempts, and record which grabbed better. One number beats an hour of arguing.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Intake mounted solidly to the chassis
- ☐ Robot still fits the legal starting size with it attached
- ☐ 10 pickup attempts counted and the result recorded

HAND OFF TO

Tell the programming crew the intake exists — PR-8 can map it to a button.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Half of all 'the robot is acting weird' problems are a low battery. A team that manages batteries wins matches that other teams lose for no visible reason.

YOU WILL FINISH WITH

A labeled, color-coded charging station and a written battery routine everyone follows.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Gather every battery in the room. Ask the coach for the team's battery sticker sheets.

15-40 min

- ☐ Number every battery and write the date it entered service on it. Apply a status sticker: charged / in use / needs charging / retired.

40-65 min

- ☐ Set up the station in ONE place: chargers labeled, a clearly marked shelf for charged batteries and a separate one for dead batteries.

65-85 min

- ☐ Measure and record each battery's resting voltage. Flag any battery that reads consistently low — it may be near the end of its life.

85-100 min

- ☐ Write the battery rules on a card and post it: who swaps, when, where dead ones go, and who is responsible for putting them on the charger.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Every battery the team owns
- Chargers
- Battery status stickers (ask the coach — the team has printable sheets)
- Multimeter or battery checker
- Permanent marker

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Start a weekly voltage log. You will spot a dying battery weeks before it costs you a match.

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Every battery numbered, dated, and stickered
- ☐ Resting voltages logged
- ☐ Station set up with charged/dead separated
- ☐ Rules card posted

HAND OFF TO

Announce the battery rules to the whole team. This only works if everyone follows it.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Getting Pollen in is one problem; putting it where you want it is a different one. Prototyping in cardboard is far faster and cheaper than discovering the flaw in aluminum.

YOU WILL FINISH WITH

A working rough prototype tested with real Pollen, with two versions and a success count for each.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Review the concept chosen in EP-4. Gather cardboard and scrap. Do not raid the good parts bin.

15-50 min

- ☐ Build the roughest version that could possibly work. Do NOT make it pretty. Speed matters far more than finish at this stage.

50-75 min

- ☐ Test with real Pollen. Do 10 attempts and count how many land where you wanted. Write the number down.

75-95 min

- ☐ Change exactly ONE thing. Test 10 more attempts. Record that number too.

95-100 min

- ☐ Photograph both versions and write down what you learned, including what surprised you.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- The chosen concept from EP-4
- Cardboard, zip ties, tape, scrap parts
- Real Pollen (10+)
- Scissors or box cutter (with supervision)

BEFORE YOU START

BA-5, EP-4

THIS UNLOCKS

IF YOU FINISH EARLY

- Sketch what the real metal version would look like and list every part you would need to order.

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ A physical prototype exists
- ☐ Two versions tested with 10 attempts each and both counts recorded
- ☐ Findings written in the notebook

HAND OFF TO

Bring the numbers to the next planning session — they decide whether this concept is worth building for real.

Who worked on this: _____ Date: _____

Robot Cart & Pit Kit

WHY THIS MATTERS

At a competition you get a small pit and very little time between matches. A team with a cart and an organized toolkit fixes the robot while other teams are still looking for a hex driver.

YOU WILL FINISH WITH

A cart the robot rides on safely and a packed pit toolkit with a checklist taped inside the lid.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Find or borrow a cart. Set the robot on it and check that it is stable and will not slide off when the cart is pushed.

20-50 min

- ☐ Pack the pit toolkit: hex drivers, spare fasteners, spare motor cables, zip ties, tape, a charged spare battery, and a multimeter if the team has one.

50-75 min

- ☐ Write the pit checklist — one line per item that must be in the box. Print it and tape it inside the lid.

75-95 min

- ☐ Dry run: unpack everything, repack it, and have a DIFFERENT person check it against the list.

95-100 min

- ☐ Park the cart where it will live. Photograph the packed kit.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ Robot rides on the cart without sliding
- ☐ Toolkit packed
- ☐ Checklist taped inside the lid and verified by a second person

WHAT YOU NEED

- A cart or rolling cabinet (ask the coach)
- A toolbox or bin
- Tools and spares
- Label tape
- Printer

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

IF YOU FINISH EARLY

- Add a laminated 5-minute repair card listing the three most likely failures and how to fix each one.

HAND OFF TO

The pit kit gets checked against its list before every event. Name who owns that.

Who worked on this: _____ Date: _____

Assemble a Programming Board

WHY THIS MATTERS

22257's first board is already built — this card is for building a SECOND one, or repairing the first. The board carries a Control Hub, a motor, a servo and three sensors, so a programming crew can work with none of the robot.

YOU WILL FINISH WITH

A fully assembled, wired programming board that powers up, with its port map taped to it.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Check with the coach first: is there a spare Control Hub for this board? Without one the board cannot run, and the existing board already has the team's second hub.
- ☐ Lay out every printed part and every component and check them against the model's parts list. One missing M4x40 stops the whole build.
- ☐ This is the REV build — use the REV cover and REV motor mounts, NOT the goBILDA ones. They are not interchangeable.

15-45 min

- ☐ Mount the Control Hub (4x M3x18mm with M3 lock nuts) and the potentiometer (4x M3x6mm).
- ☐ These threads go into printed plastic. Snug them and STOP — overtightening strips the part and you will have to reprint it.

45-70 min

- ☐ Mount the servo (4x M4x40mm), the touch sensor (2x M4x12mm) and the color sensor (2x M4x12mm).
- ☐ Mount the motor: front mount (2x M4x14mm to the board, 4x M3x8mm to the motor), then the rear support and clamp (2x M4x40mm).
- ☐ Fit the motor wheel and press on the potentiometer knob.

70-90 min

- ☐ Wire it: motor to a motor port, servo to a servo port, touch sensor and potentiometer to their ports, color sensor to an I2C port.
- ☐ WRITE DOWN which port every single thing went to. The next crew needs this list for PR-2 and cannot guess it.

90-100 min

- ☐ Power it from a charged 12V battery and confirm the Control Hub lights come up. Tape the port map to the board.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Every component mounted, nothing loose, no stripped printed parts
- ☐ Port map written and taped to the board
- ☐ Hub powers up on battery

WHAT YOU NEED

- A spare REV Control Hub — confirm with the coach BEFORE starting; the existing board already has the team's second hub
- Printed parts, REV variant: board, cover, front motor mount, rear support, rear clamp, motor wheel, potentiometer knob
- REV components: HD Hex Motor 40:1 (41-1301), Smart Robot Servo (41-1097), Touch Sensor (41-1425), Color Sensor V3 (31-1557), Potentiometer (31-1155)
- M3 and M4 hardware plus nylon lock nuts — see the model's parts list
- Hex drivers and a charged 12V battery

BEFORE YOU START

EP-7

THIS UNLOCKS

—

IF YOU FINISH EARLY

- If the team has a second Control Hub and a second set of printed parts, build a second board. Two boards means two programming crews at once.

HAND OFF TO

Give the port map to whoever takes PR-2. Cards PR-10, PR-11 and PR-12 all need this board.

Who worked on this: _____ Date: _____

Install the Pinpoint Odometry Pods

Robot

Some experience

2-3 people

100 min

WHY THIS MATTERS

Odometry is how the robot knows where it is. Two unpowered tracking wheels plus the Pinpoint's pre-calibrated IMU give a position good enough to follow a real path, instead of guessing with timers and hoping the battery holds up.

YOU WILL FINISH WITH

Both pods and the Pinpoint mounted and wired, status LED green, and the pod offsets measured.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Read the setup section of the goBILDA Pinpoint user guide.
- ☐ Mount the Pinpoint sticker-side up, somewhere you can actually see its status LED without disassembling anything.

15-50 min

- ☐ Mount the pods on the adapter: ONE tracking forwards (the X pod) and ONE tracking sideways (the Y pod).
- ☐ Both wheels must sit flat on the floor with the 4-bar free to swing. Lift each pod wheel by hand and let go — it should spring back down to the floor every time.

50-70 min

- ☐ Wire it. Forward pod into the Pinpoint's X port, strafe pod into the Y port. Getting these two swapped causes hours of confusion later, so check twice.
- ☐ Pinpoint into an I2C port on the Control Hub — but NOT I2C port 0. Pedro Pathing needs port 0 left alone.

70-85 min

- ☐ Power up and read the status LED. Green is ready. Purple means no pods detected at all, blue means no X pod, orange means no Y pod.

85-100 min

- ☐ MEASURE THE POD OFFSETS now, while you can still reach them.
- ☐ X Pod Offset is how far sideways the X pod sits from the robot's tracking point — left of the point is positive, right is negative.
- ☐ Y Pod Offset is how far forwards the Y pod sits — forwards is positive, backwards is negative.
- ☐ Record both in MILLIMETRES in the notebook. PR-15 needs these exact numbers.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ Both pod wheels sit flat on the floor and spring back when lifted
- ☐ X pod in the X port, Y pod in the Y port
- ☐ Pinpoint on an I2C port that is not port 0
- ☐ Status LED green
- ☐ Both offsets measured in mm and written in the notebook

WHAT YOU NEED

- 4-Bar Odometry Pack (goBILDA 3203-3110-0002)
- The printed adapter from CD-8
- Robot with a wired Control Hub (BA-4)
- Hex drivers
- The Pinpoint user guide
- Ruler or calipers

BEFORE YOU START

BA-4, CD-8

THIS UNLOCKS

PR-15

IF YOU FINISH EARLY

- Push the robot a full metre by hand and watch the pod wheels. If either skips, slips or lifts, fix the mounting now — odometry that slips is worse than no odometry, because the code believes it.

HAND OFF TO

The two offsets go straight to whoever takes PR-15.

Who worked on this: _____ Date: _____

Starter Bot: Parts Check & Cut List

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

REV's guide assumes a complete Starter Kit V3.1 sitting on the bench. We have the parts from an earlier season's Starter Kit plus the team's own parts library, so job one is finding out exactly what we have and what we are short. Job two is every cut on the list, because those four cuts feed four subassemblies that get built in parallel. One missing part, or one wrong cut, stops three crews.

YOU WILL FINISH WITH

Every part checked against the bill of materials, every piece on the cut list cut, labelled and bagged by assembly.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-25 min

- ☐ Walk the bill of materials against the parts library line by line. Tick what we have, circle what we are short. The short list goes on the whiteboard today and gets ordered the same day — a missing four-dollar part stops the build for a week.

25-35 min

- ☐ Cutting rules. REV's guide asks for a hacksaw or band saw; we use the chop saw and it works, so the rules are ours: clamp every piece, never hold it by hand, let the blade reach speed before it touches metal, and cut slowly. Only students signed off in EP-7 run the saw.

30-75 min

- ☐ Cut the 15mm extrusion. From 420mm stock: 323 x2, 278.5 x2, 248 x3, 83 x3, 70 x4, 63 x1. From 225mm stock: 218 x3. Measure twice, deburr every cut.

75-90 min

- ☐ Cut two 308mm and two 252mm hex shafts. Cut the polycarb with tin snips or the band saw: 329x84 launcher floor, 248x112 intake floor, 248x64 intake lid, 200x72 intake side. Make up the 54-link chain.

90-100 min

- ☐ Label every piece with its length and its assembly. Bag by assembly: Launcher Floor, Intake, Flywheel & Feeder, Pollen Pusher.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Every BOM line checked off or flagged as missing
- ☐ Every cut made, deburred and within 1mm
- ☐ Four labelled bags on the shelf

WHAT YOU NEED

- Build guide, BOM on page 2 and cut list on page 3
- The team parts library and the old Starter Kit
- Chop saw (or hacksaw), file, calipers
- Tin snips or the scroll saw for polycarb
- Tape and labels, sandwich bags
- Safety glasses

LINKS

Clickable on the Coaches Corner site

- **REV Starter Bot build guide (PDF)**

www.revrobotics.com/content/technical-resources/.../2026-27_REV_DUO_FTC_Starter_Bot-Build_Guide_1567.pdf

BEFORE YOU START

BA-1, EP-7

THIS UNLOCKS

BA-12, BA-13, BA-14, BA-15

IF YOU FINISH EARLY

- Pre-load the brackets while you are at it — REV's guide needs 28 ninety-degree, 8 inside corner and 8 lap corner brackets loaded with a screw and a nyloc in every hole. It is the best parallel job in the shop.

HAND OFF TO

The four bags go to BA-12, BA-13, BA-14 and BA-15, which can all run in the same meeting.

Who worked on this: _____ Date: _____

Starter Bot: Launcher Floor

WHY THIS MATTERS

The launcher floor is the ramp Pollen rides from the intake up to the flywheel, and it carries the intake floor too. It is a bench assembly — it does not need the robot, so it runs at the same time as the other three.

YOU WILL FINISH WITH

A finished, square launcher floor assembly on the shelf.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Read pages 87 to 96 all the way through before picking up a screw. Lay the parts out in the order the guide calls for them.

15-25 min

- ☐ Pre-load the brackets the guide asks for — a screw and a nyloc finger-started in each hole, ribbed side away from the screw head. Pre-loading is half of every REV build.

25-55 min

- ☐ Build the extrusion frame: two 278.5mm, two 218mm, two 83mm. Brackets finger-tight only until the whole frame exists.

55-80 min

- ☐ Fit both polycarb pieces. Polycarb cracks if you overtighten — snug is enough, and a cracked piece is a reorder and a lost week.

80-95 min

- ☐ Square it up, check that the two diagonals match, then tighten in a cross pattern.

95-100 min

- ☐ Photograph it, label it LAUNCHER FLOOR, shelve it.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Build guide pages 87-96
- The Launcher Floor bag from BA-11
- Launcher floor and intake floor polycarb
- 5.5mm wrench and nut driver, hex drivers
- A square

BEFORE YOU START

BA-11

THIS UNLOCKS

BA-16

IF YOU FINISH EARLY

- Mark on the drawing where a different floor angle could bolt on. Mk-2 will want to change the launch angle and you will be glad the holes exist.

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Matches the photo at the end of the section
- ☐ Diagonals within 2mm of each other
- ☐ No cracks or stars around any polycarb hole
- ☐ Labelled and on the shelf

HAND OFF TO

Goes to BA-16 for final assembly. Do not let it get used as a shelf in the meantime.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Eight rubber flaps on a chain-driven shaft are what sweep Pollen off the floor and into the robot. It is the simplest mechanism on the Starter Bot and the one that decides whether you ever get to shoot.

YOU WILL FINISH WITH

A finished intake assembly whose shafts spin freely and together by hand.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Read pages 97 to 104 through. Count flaps, spacers and sprockets against the parts list before you start.

15-40 min

- ☐ Build the two 308mm flap shafts — flap, spacer, flap, spacer, all the way along. Get the order right first time; taking one apart is a twenty minute penalty.

40-65 min

- ☐ Build the 248mm extrusion frame and the two 225mm side rails with their motion and rod end brackets.

65-85 min

- ☐ Mount the Core Hex motor and loop the 54-link chain over the sprockets. Slide the shafts apart until the chain is tight, then tighten just enough to hold it.

85-95 min

- ☐ Fit the intake lid. Turn a shaft by hand — it should spin freely and both shafts should turn together.

95-100 min

- ☐ Photograph, label INTAKE, shelve.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Both flap shafts spin freely and turn together
- ☐ Chain tight with no slack loop
- ☐ Lid fitted with no cracked polycarb
- ☐ Labelled and on the shelf

WHAT YOU NEED

- Build guide pages 97-104
- The Intake bag from BA-11
- 8 intake flaps, Core Hex motor, 54-link chain
- Two 308mm hex shafts
- Intake lid polycarb
- Hand tools

BEFORE YOU START

BA-11

THIS UNLOCKS

BA-16

IF YOU FINISH EARLY

- Run it on the bench with the 1x30A Motor Commander and roll a Pollen at it. Finding out now that the flaps are too soft is worth a great deal later.

HAND OFF TO

Goes to BA-16. Tell that crew the chain tension is not final — it gets set again once the intake is on the robot.

Who worked on this: _____ Date: _____

Starter Bot: Launcher Flywheel & Feeder

No hardware

Challenge

2-3 people

100 min

WHY THIS MATTERS

This is the mechanism that scores. Two 90mm grip wheels on a 1:1 gearbox spin fast enough to throw a Pollen into the Hive, and a geared feeder roller pushes them in one at a time. It is also the most dangerous thing on the robot — treat a spun-up flywheel like a saw blade.

YOU WILL FINISH WITH

The launcher and feeder assembly finished, spinning true, on the shelf.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Read pages 105 to 119 through. Note that this gearbox is 1:1 — no cartridges go in it. It is the only 1:1 on the robot and it is the easiest thing here to build wrong.

15-45 min

- ☐ Flywheel shaft: collar, two 90 tooth gears, two 90mm grip wheels, two more 90 tooth gears, collar. Leave about 30mm of shaft for the bearing. Spin it — fix any wobble now.

45-70 min

- ☐ Feeder shaft: collar, four 1in soft grip wheels, collar, then the bearing blocks, and the 60 tooth gear on the other end.

70-90 min

- ☐ Build the feeder servo assembly, then join flywheel, feeder and servo onto the 323mm and 218mm extrusion. Check the 90 and 60 tooth gears mesh without binding.

90-100 min

- ☐ Turn the flywheel by hand and watch the feeder turn with it. Photograph, label, shelf.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Build guide pages 105-119
- The Flywheel & Feeder bag from BA-11
- UltraPlanetary gearbox and HD Hex motor
- Four 90 tooth gears, one 60 tooth gear
- Two 90mm grip wheels, four 1in soft grip wheels
- Two 252mm hex shafts, a Smart Robot Servo

BEFORE YOU START

BA-11

THIS UNLOCKS

BA-16

IF YOU FINISH EARLY

- Put a stripe of tape on the flywheel shaft. At full speed you cannot see it turning, and the stripe tells you instantly whether it is spun up.

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ 1:1 gearbox confirmed — no cartridges installed
- ☐ Flywheel spins with no visible wobble
- ☐ Gears mesh and turn freely by hand
- ☐ Labelled and on the shelf

HAND OFF TO

Goes to BA-16, then to PR-20 for tuning. Safety glasses whenever this runs, every time, no exceptions.

Who worked on this: _____ Date: _____

Starter Bot: Pollen Pusher

WHY THIS MATTERS

The pusher is a servo-driven arm on the side of the robot that combs Pollen out of the Flowers. It is the smallest assembly on the Starter Bot and the best first real build job for someone who has not done much yet.

YOU WILL FINISH WITH

A finished pollen pusher that sweeps its full travel without hitting anything.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Read pages 41 to 49. This assembly uses 30 degree brackets — do not substitute 90s because they look similar in the bag.

15-40 min

- ☐ Build the extrusion arm from the 63mm, 83mm and 120mm pieces with their brackets.

40-70 min

- ☐ Centre the servo BEFORE attaching the arm. A servo fitted at the end of its travel can only move one way, and you will not find that out until the robot is built.

70-90 min

- ☐ Fit the aluminium double servo arm and tighten both nuts, rotating the servo to reach each one in turn.

90-100 min

- ☐ Sweep it through its travel by hand. Photograph, label, shelf.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN – ALL 4 MUST BE TRUE

- ☐ Servo centred before the arm went on
- ☐ Arm sweeps its full travel without binding
- ☐ Both servo arm nuts tight
- ☐ Labelled and on the shelf

WHAT YOU NEED

- Build guide pages 41-49
- The Pollen Pusher bag from BA-11
- Smart Robot Servo V2 Balanced
- Aluminium double servo arm
- 63mm, 83mm and 120mm extrusion
- Hand tools

BEFORE YOU START

BA-11

THIS UNLOCKS

BA-16

IF YOU FINISH EARLY

- Work out and write down the two servo positions the code will need — up and down. PR-19 needs those numbers and guessing wastes a meeting.

HAND OFF TO

Goes to BA-16. Give the up and down positions to the programming crew.

Who worked on this: _____ Date: _____

Mount the Starter Bot on Our Mecanum Chassis

WHY THIS MATTERS

This is the part REV has no guide for. Their superstructure bolts to a two-motor channel drivetrain; ours has to bolt to the mecanum chassis we built in September. The good news is both use the same 408mm and 248mm C channel, so most of the interface transfers. The rest is our engineering — and it is exactly what judges ask about.

YOU WILL FINISH WITH

Launcher, feeder, intake and pollen pusher mounted on the mecanum chassis, square, with nothing rubbing.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Lay the subassemblies on the chassis roughly where they go and photograph it. Check the CD-9 drawing against reality before anything is tightened.

20-45 min

- ☐ Fit the four 70mm uprights and the two 248mm cross rails to the drive rails. These are what the launcher hangs from — square and evenly spaced, or the shot drifts.

45-70 min

- ☐ Bolt the launcher and feeder assembly onto the uprights, then fold the launcher floor back onto it and secure it.

70-90 min

- ☐ Hang the intake so its lid sits just above the feeder wheels. Our wheels are 15mm smaller in diameter than REV's, so the frame sits 7.5mm lower and every height measured from the floor has changed. Fit the pollen pusher on the side.

90-100 min

- ☐ Turn every wheel and roller by hand. Anything that rubs gets fixed now. Photograph the robot from four sides.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ All four subassemblies mounted and square
- ☐ Intake lid clears the feeder wheels
- ☐ Nothing rubs when wheels and rollers are turned by hand
- ☐ Odometry pods still reach the floor
- ☐ Photographed from four sides

WHAT YOU NEED

- Mecanum chassis (BA-3, BA-4)
- The four subassemblies
- The marked-up drawing from CD-9
- Four 70mm and two 248mm extrusion, brackets, U channel endcaps
- Hand tools

BEFORE YOU START

BA-12, BA-13, BA-14, BA-15, CD-9

THIS UNLOCKS

BA-17

IF YOU FINISH EARLY

- Write down every place our mounting differs from REV's guide. That list is a portfolio page and a judging answer in one.

HAND OFF TO

Do not wire anything yet — BA-17 takes it from here.

Who worked on this: _____ Date: _____

Add the Expansion Hub: Power, Comms & Six Motors

Robot

Challenge

2-3 people

100 min

WHY THIS MATTERS

A Control Hub has four motor ports. Mecanum uses all four. The intake and the flywheel are motors five and six, so this robot cannot work without an Expansion Hub. Ours is the one from last season that got a reputation for dropping out, so this card has a second job: prove it or condemn it. An intermittent hub is the worst thing on a robot, because it looks exactly like a software bug and it fails in the middle of a match.

YOU WILL FINISH WITH

An Expansion Hub mounted, powered, talking to the Control Hub, with the intake and flywheel on it.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Update the Expansion Hub's firmware in the REV Hardware Client before anything else — stale firmware is the most common cause of a hub that 'goes flaky'.

15-35 min

- ☐ Find it a home, then bolt it down: reachable, off the floor, away from the flywheel, LED visible. Zip ties alone will not survive a match. Power it with the XT30 extension so both hubs draw from one battery through one switch.

35-50 min

- ☐ Comms: the 3-pin cable between the two hubs. Power up and watch the Expansion Hub's LED settle into a steady pattern, not an error blink.

50-75 min

- ☐ Prove the hub. Run every motor port one at a time, then wiggle each cable while it runs. Write down which ports are good. We only need two, so one bad port is survivable — a hub that drops the whole link is not.

75-100 min

- ☐ Wire the intake and the flywheel to two good ports, routed away from anything that turns. Strain-relieve and cable-manage the whole robot, tug every connector, photograph it for the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Firmware updated; every port tested and the good ones written down
- ☐ Link survives wiggling every cable while motors run
- ☐ Both hubs powered from one battery through the main switch
- ☐ Intake and flywheel wired and strain-relieved, nothing reaching a moving part

WHAT YOU NEED

- Expansion Hub with its XT30 extension and 3-pin communication cable
- Mounting hardware
- The robot
- Zip ties and wire cutters

BEFORE YOU START

BA-16, BA-4

THIS UNLOCKS

BA-18, PR-18

IF YOU FINISH EARLY

- Label both ends of every cable. When something fails at an event, labels are the difference between two minutes and twenty.

HAND OFF TO

Hand it to PR-18 to configure. Nobody writes code until that LED is healthy. If the link dropped even once during the test, tell the coach today — that is a purchase decision, not a debugging job.

Who worked on this: _____ Date: _____

Inspection Dry Run: Make It Legal

Robot

Some experience

3-4 people

100 min

WHY THIS MATTERS

A robot that fails inspection does not play. Rookie teams get caught by the same handful of things: too big at the start, exposed wiring, no team number visible, the wrong battery. Finding them in the shop costs an hour. Finding them at the event costs your first two matches.

YOU WILL FINISH WITH

The robot passes our own inspection against the real checklist, a week before the event.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Print the inspection checklist and read it top to bottom as a team before measuring anything. Most of it is about things you can still fix.

20-45 min

- ☐ Size it. Tape the legal starting outline on the bench and put the robot inside it in its starting configuration. If it does not fit, that is today's job and nothing else matters.

45-70 min

- ☐ Electrical: approved battery, main switch reachable, no exposed conductors, nothing pinched, both hubs properly fastened.

70-85 min

- ☐ Identification: team number visible from the required sides at the required size. REV's sign holders take a 7 by 3 inch sign — make ours now if they do not exist.

85-100 min

- ☐ Walk the whole checklist and write PASS or FAIL beside every line. Fix what you can today; put a name and a date against everything left.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- The official inspection checklist
- Competition Manual
- Tape measure and a taped starting outline
- The robot
- Card and markers for team number signs

BEFORE YOU START

BA-17

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Have a student who did not build the robot run the inspection. Fresh eyes find what the builders have stopped seeing.

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Every checklist line marked PASS or FAIL
- ☐ Robot fits inside the legal starting outline
- ☐ Team number signs made and mounted
- ☐ The fail list has an owner and a date before the event

HAND OFF TO

The marked checklist goes in the pit binder (EP-10). Re-run it after any change to the robot.

Who worked on this: _____ Date: _____

MG-1

MERCH & GIVE-AWAY

Give-Away Plan & Budget

No hardware

Start here

2-3 people

100 min

WHY THIS MATTERS

Trading give-aways with other teams is one of the best parts of a competition and it is how people remember you afterward. But it costs real money, so decide before you buy.

YOU WILL FINISH WITH

One chosen give-away with a real cost per unit, a justified quantity, and a total that fits the budget.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Get the give-away budget from the coach. Write it on the board where everyone can see it.

15-40 min

- ☐ Brainstorm at least 8 options: buttons and pins, stickers, 3D-printed tokens, keychains, trading cards, bracelets, patches, magnets.

40-65 min

- ☐ For your top 3, find REAL prices online, or calculate material cost if you are making them. Work out cost per unit and total.

65-85 min

- ☐ Justify the quantity: number of teams at your events multiplied by 2, plus extras for judges, sponsors, and volunteers. Show the math on the board.

85-100 min

- ☐ Pick one primary and one backup. Write a one-paragraph recommendation to the coach with the numbers in it.
- ☐ Then price the team's one embroidered award patch: FULL STACK, 10 pieces, 2 inch, iron-on backing. Get THREE quotes — the same patch varies by about three times between vendors. Include Ninja Transfers' heat-pressable patches, since Herobotix already buys from them and owns a press.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- The give-away budget number (ask the coach)
- Laptop for pricing
- Whiteboard
- Calculator

BEFORE YOU START

EP-1

THIS UNLOCKS

MG-2, MG-5

IF YOU FINISH EARLY

- Price a second, nicer item to hand specifically to sponsors and judges.

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ One give-away chosen with a real cost per unit
- ☐ Quantity justified with visible math, total under budget
- ☐ Three comparable quotes for the embroidered FULL STACK patch
- ☐ Written recommendation handed to the coach

HAND OFF TO

The coach approves the spend. Once approved, MG-2 designs the art.

Who worked on this: _____ Date: _____

MG-2

MERCH & GIVE-AWAY

Design the Give-Away Art

No hardware

Some experience

2-4 people

100 min

WHY THIS MATTERS

The art carries the team name to people who have never met you. It has to be readable from three feet away in about one second.

YOU WILL FINISH WITH

Print-ready artwork at the correct size and resolution, saved in the team's shared folder.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Pull up the identity brief and logo sketches from EP-1. Get the exact dimensions of the item from MG-1.

15-50 min

- ☐ Draft 3 versions. Keep it simple — a button is small and a sticker gets looked at for one second. Team name AND team number 22257 must both appear.

50-70 min

- ☐ The squint test: print a draft at actual size, stand 6 feet back. Can you still read the name and number? If not, make them bigger and cut something else.

70-90 min

- ☐ Show it to someone not on the crew. Ask what they noticed first. Revise once based on what they say.

90-100 min

- ☐ Export at the right physical size and 300 dpi for print. Save to the team shared folder with a clear filename including the date.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Identity brief and logo sketches (EP-1)
- The exact size and shape of the item (MG-1)
- Design software or Canva
- A printer for test prints

BEFORE YOU START

EP-1, MG-1

THIS UNLOCKS

MG-3

IF YOU FINISH EARLY

- Make a matching version sized for a sticker and another sized for a social media post.

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Passes the 6-foot squint test
- ☐ Team name and number 22257 both present and legible
- ☐ Exported at correct size and 300 dpi
- ☐ Saved in the shared folder with a dated filename

HAND OFF TO

Give the file to MG-3 for production, and to the MF-4 social crew to reuse.

Who worked on this: _____ Date: _____

Produce the Give-Aways (assembly line)

WHY THIS MATTERS

Making 200 of something is a completely different job from making one. Run it like a production line and it is fun; run it randomly and it is miserable.

YOU WILL FINISH WITH

A counted, bagged, quality-checked batch stored with the competition gear.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Set up stations. Assign roles: cut, assemble, quality check, count and bag.

15-25 min

- ☐ Make ONE complete unit together. Everyone looks at it. Agree out loud that this is the standard.

25-80 min

- ☐ Run the line. Rotate stations halfway through so nobody is stuck on one job for an hour.

80-90 min

- ☐ Quality check: pull 10 at random and inspect them. If there are rejects, fix the PROCESS, not just those pieces.

90-100 min

- ☐ Count the final total, bag them, write the count on the bag, and store it with the pit kit.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Approved art (MG-2)
- Blanks and supplies
- Button press / cutter / printer as needed
- Zip bags
- Marker

BEFORE YOU START

MG-2

THIS UNLOCKS

IF YOU FINISH EARLY

- Record how many you produced per minute. You will want that number the next time you plan a batch.

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ Batch counted and bagged with the count written on the bag
- ☐ Random sample of 10 passed inspection
- ☐ Stored with the competition gear

HAND OFF TO

Tell the coach the count. Add the bag to the pit kit checklist (BA-8).

Who worked on this: _____ Date: _____

MG-4**MERCH & GIVE-AWAY**

Team Card / Business Card

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

Judges, sponsors, and other teams all ask 'how do I find you?'. A card answers that in one hand-off and it stays in their pocket.

YOU WILL FINISH WITH

A proofread, print-ready card design with a known print source and cost.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Decide the content: team number 22257, team name, school, season, a photo or logo, and ONE contact.
- ☐ IMPORTANT: use a team email or the coach's contact. Never a student's personal phone, email, or social handle.

15-45 min

- ☐ Lay out front and back at standard card size, 3.5 x 2 inches. Keep everything at least 1/8 inch inside the edge so nothing gets trimmed off.

45-65 min

- ☐ Proofread it. Then have two other people proofread it. A typo on 200 printed cards is permanent.

65-85 min

- ☐ Print a test page on plain paper, cut one out, and hold it in your hand. Fix whatever looks wrong at real size.

85-100 min

- ☐ Save the print-ready file and write down where you would print it and what it costs.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Proofread by 3 different people
- ☐ Test print cut out and checked at actual size
- ☐ No student personal contact info anywhere on it
- ☐ Print source and cost written down

WHAT YOU NEED

- Identity brief and logo (EP-1)
- Team photo
- Design software
- Printer for a test page
- Scissors

BEFORE YOU START

EP-1

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Add a QR code pointing at the team's page on the Herobotix website.

HAND OFF TO

Coach approves before printing. Cards go in the pit kit for competitions.

Who worked on this: _____ Date: _____

3D-Printed Give-Away

WHY THIS MATTERS

A printed item is cheap per unit and unmistakably yours. It is also real CAD practice that transfers directly to designing robot parts.

YOU WILL FINISH WITH

One test print in hand and a realistic run plan with a start date.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Confirm printer access and which filament colors are available. Get the coach's OK on how much filament you may use.

15-50 min

- ☐ Model or modify a simple item in Onshape: a keychain, badge, or Pollen-themed token. Emboss 22257 into it so it is unmistakably yours.

50-70 min

- ☐ Slice it and read the estimated print time. Multiply by how many you need. Is that actually possible before the event? If not, shrink it or simplify it until it is.

70-90 min

- ☐ Print ONE. Inspect it: does the number read clearly? Does it survive being dropped on the floor?

95-100 min

- ☐ Write the run plan: units per plate, hours per plate, plates needed, and the date printing must START.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- 3D printer access (confirm with the coach)
- Filament
- Onshape or slicer software
- Calipers or ruler

BEFORE YOU START

MG-1

THIS UNLOCKS

IF YOU FINISH EARLY

- Design a version that snaps onto a lanyard or a backpack zipper — people keep those.

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ One test print in hand and inspected
- ☐ Print time math done and shown
- ☐ Run plan written with a start date

HAND OFF TO

Give the start date to the coach. If it is soon, start printing.

Who worked on this: _____ Date: _____

Design the Award Stickers

WHY THIS MATTERS

Finishing cards earns awards. Ten of the eleven are stickers the team prints itself on full-sheet label paper, so each one costs a few cents and a lost one is simply reprinted — which means the design, not the budget, decides whether people actually want them. The eleventh, FULL STACK, is a real embroidered patch for the team shirt, and embroidery is far less forgiving than a printer.

YOU WILL FINISH WITH

One shared sticker frame plus seven category marks, print-tested on the label paper, and a FULL STACK design simple enough to embroider.

WHAT YOU NEED

- Identity brief and logo from EP-1
- The team awards sheet, so you know how many designs are needed
- Paper, pencils and a ruler
- Design software
- Plain paper, the team's label paper, scissors

BEFORE YOU START

EP-1

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Design the three special awards too — Assist, First Light and Keystone. They are stickers like the category ones, so each extra design costs essentially nothing to print.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Read the awards sheet and count what you must design: one shared frame plus seven category marks. Get the team logo and colours from EP-1.

15-35 min

- ☐ Design the FRAME first — the shape every sticker shares. Make it 2 inches across and pick a shape that is easy to cut with scissors: a circle or a rounded square, not a star.
- ☐ The team name and number live on the frame, so you draw them once instead of seven times.

35-65 min

- ☐ Design the seven category marks that drop into the frame. One clear idea each, drawn thick — a mark has to read at two inches, so one object works and a whole scene does not.

65-80 min

- ☐ THE RULES CHECK for every sticker: no small text, nothing important within 1/8 inch of the edge (that is where scissors wander), and it still reads at 2 inches.
- ☐ FULL STACK has stricter rules because it is being embroidered: no more than 4 colours, no line thinner than 1/16 inch, no gradients, no fine detail. Embroidery is thread — a shape smaller than a stitch simply will not exist.

80-100 min

- ☐ Print on PLAIN paper first and check at actual size; only then print one sheet on the label paper. Cut a few out and fix whatever fails at two inches.
- ☐ Hand the FULL STACK artwork to whoever has MG-1 for quoting.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ One shared frame carrying the team name and number, plus seven marks that drop into it
- ☐ Sticker rules passed on all designs, embroidery rules passed on FULL STACK
- ☐ A test sheet printed on the label paper, cut out, and checked at actual size

HAND OFF TO

Save the print-ready sheet in the shared folder so anyone can reprint a lost sticker. FULL STACK artwork goes to MG-1 for quoting.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

The shirt is what the team looks like to everyone at an event. Herobotix shirts are made in-house: plain shirts, full-colour DTF transfers ordered by size, and the club's own heat press. That changes the rules — colour is free, but every square inch of graphic costs money.

YOU WILL FINISH WITH

Three or more concepts drawn front and back, every graphic sized in inches, presented, with written feedback collected.

WHAT YOU NEED

- Printed blank T-shirt outlines, front and back
- Coloured markers and a ruler
- Identity brief (EP-1)
- Sticky notes

BEFORE YOU START

EP-1

THIS UNLOCKS

TS-2

IF YOU FINISH EARLY

- Price your own concept: add up the square inches of all its graphics. Two designs that look equally good can cost very different amounts to print.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Print blank T-shirt outlines and review the EP-1 identity brief. Remember how ours are made: transfers print in FULL COLOUR and are priced by SIZE, so extra colours cost nothing and bigger graphics do.

15-25 min

- ☐ Get the rules from the coach and write them on the board: must the school or Herobotix logo appear, and what is the budget per shirt.
- ☐ Keep every single graphic 10 inches wide or less — that is the largest transfer the club press has handled.

25-35 min

- ☐ RESERVE TWO SPACES on every blank before designing: a 2.5 inch patch spot (sleeve or upper back) for FULL STACK, and a sponsor area — last season every Herobotix shirt got four sponsor strips about 4.6 inches wide. Ask the coach if sponsors return this year.

35-70 min

- ☐ Split into small groups. Each group designs one complete concept, front and back, keeping both reserved spaces clear. Team number 22257 must be readable from across a gym.
- ☐ Write each graphic's width and height in inches right on the drawing — that is literally what the transfer is priced on.

70-90 min

- ☐ Post every concept. Each group presents theirs for 2 minutes.

90-100 min

- ☐ Collect written feedback on sticky notes stuck to each design. Do NOT vote today — that is TS-2, and sleeping on it produces better decisions.

NOTES, SKETCHES, WHAT SURPRISED YOU**DONE WHEN — ALL 4 MUST BE TRUE**

- ☐ 3+ concepts drawn front and back, every graphic 10 inches wide or less
- ☐ Each graphic's size written on the drawing in inches
- ☐ Patch spot and sponsor area kept clear on every concept
- ☐ Every concept presented and sticky-note feedback collected

HAND OFF TO

Leave the designs and feedback up on the wall for TS-2.

Who worked on this: _____ Date: _____

Vote & Make the Art Print-Ready

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

A transfer company prints exactly the file you upload, at exactly the size you order. This is where a marker sketch becomes files that can be printed and pressed without surprises.

YOU WILL FINISH WITH

The winning design as one print-ready file per transfer, at actual size, coach-approved and saved with a version number.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Run the vote. Every member gets one vote. The coach holds a veto only on rule and budget issues. Record the result in the notebook.

20-55 min

- ☐ Rebuild the winner as clean digital art AT ACTUAL PRINT SIZE, in inches. DTF prints full colour, so colour is not your limit — but hairline strokes and tiny text still disappear on fabric. Keep lines bold.

55-80 min

- ☐ Split it into one file per transfer (front, back, sleeve...). Each on a TRANSPARENT background — anything that is not transparent gets printed, including a white box around your logo.
- ☐ Export at high resolution; transfer companies typically want PNG at 300 DPI at the real size. Check the vendor's upload rules before you export.
- ☐ Confirm the patch spot and sponsor area from TS-1 are still clear, and write down where each transfer goes on the shirt, measured from the collar.

80-95 min

- ☐ Get coach approval. Make any required changes right now while everyone is here.

95-100 min

- ☐ Save the files into the shared folder with a version number and the date in each filename.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Vote recorded
- ☐ One file per transfer at actual size, transparent background, high resolution
- ☐ Placement of each transfer written down, and the patch spot and sponsor area still clear
- ☐ Coach approved and saved with version number and date

WHAT YOU NEED

- TS-1 concepts and feedback
- Design software (Inkscape, Illustrator, or Canva)
- Laptop
- A ruler, to check sizes on a real shirt
- Coach available to approve

BEFORE YOU START

TS-1

THIS UNLOCKS

TS-4

IF YOU FINISH EARLY

- Print each file on plain paper at 100% scale, cut it out and lay it on a real shirt. Nothing catches a wrong size faster.

HAND OFF TO

The files and the transfer sizes go to TS-4 for pricing. The placement notes go to TS-5.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

The single most common shirt disaster is wrong sizes. Collect them properly, in person, once. This can run in parallel while others design.

YOU WILL FINISH WITH

A complete size count covering every member, mentor, and any parents ordering, with spares added.

WHAT YOU NEED

- Paper form or a simple online form
- A sample shirt in a couple of sizes if you can get one
- Access to the team's Band group for stragglers

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

TS-4

IF YOU FINISH EARLY

- Track who has paid, so the treasurer is not chasing it in three weeks.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-20 min**

- ☐ Build a simple form: name, role (student / mentor / parent), size, adult or youth, quantity, paid yes/no.

20-40 min

- ☐ Bring an actual sample shirt if possible. People guess wrong from a size chart and guess right when they hold one.

40-70 min

- ☐ Collect from everyone present today. Then list who is missing and message them through the team Band group.

70-90 min

- ☐ Total it into an order sheet: sizes across the top, quantities down. Add 2-3 spares in the most common sizes.

90-100 min

- ☐ Hand the totals to the coach along with a hard deadline for anyone who has not responded.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Every present member has a size recorded
- ☐ Missing people contacted through Band
- ☐ Totals sheet complete with spares
- ☐ Deadline set and communicated

HAND OFF TO

TS-4 needs these quantities to get comparable quotes.

Who worked on this: _____ Date: _____

Price the Shirts: Blanks + Transfers

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

Our shirts cost a blank shirt plus the transfers pressed onto it. Last season Beanie Bots made theirs exactly this way: three graphics a shirt came to about \$7 of transfers per shirt, shipping and tax included. Pricing both halves properly is a real business skill, and it is judged.

YOU WILL FINISH WITH

A per-shirt cost table built from real quotes for blanks and transfers, plus a recommendation with an order-by date.

WHAT YOU NEED

- Print-ready files and transfer sizes (TS-2)
- Size totals (TS-3)
- Laptop
- Coach approval before contacting anyone

BEFORE YOU START

TS-2, TS-3

THIS UNLOCKS

TS-5

IF YOU FINISH EARLY

- Ask the coach whether the other Herobotix teams are ordering transfers too. Combining orders can cross the 50-transfer discount for everybody.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Write down the exact spec: quantity by size from TS-3, every transfer's width and height from TS-2, and the date the shirts must be in hand.
- ☐ Get the coach's OK before contacting any business or starting any cart.

20-50 min

- ☐ Price BLANK shirts from three places for the same kind of shirt — same fabric, same weight. Cotton or a cotton blend; check the transfer company says it works on that fabric.

50-75 min

- ☐ Price the TRANSFERS at your exact sizes. Start with Ninja Transfers, which Herobotix used last season, and compare one other DTF company.
- ☐ Note the quantity discounts — last year's orders got 20% off at 15+ transfers and 30% off at 50+.
- ☐ Add one or two SPARE transfers of each graphic to the main order. A two-transfer reorder last season cost almost \$10 a transfer once shipping was added.

75-90 min

- ☐ Build the table: blank shirt + transfers + shipping + tax, per shirt and in total.

90-100 min

- ☐ Write a recommendation with a reason. Work backward from the event: transfers take days to ship and pressing needs a meeting, so put a bold ORDER BY date at least two weeks before the shirts are needed.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Blank shirt prices from 3 sources for the same kind of shirt
- ☐ Transfer prices from 2 companies at the exact sizes, with spares included
- ☐ Per-shirt table including shipping and tax
- ☐ Recommendation with an order-by date

HAND OFF TO

The coach places the order. When the transfers and blanks arrive, TS-5 presses them.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

This is where a stack of blank shirts becomes the team's look — or gets ruined one shirt at a time. The heat press runs above 300 degrees, so an adult runs it or supervises one student at a time, and a practice shirt always goes first.

YOU WILL FINISH WITH

Every shirt on the order sheet pressed, checked, counted and labelled by size.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Adult sets up the press and heats it to 310-320 F — but read the instructions that came with the transfers first, because settings vary by company and fabric.
- ☐ Make a placement guide from the TS-2 notes, like 'front graphic top edge 3 inches below the collar', and tape it next to the press.

15-30 min

- ☐ **TEST ON THE SPARE SHIRT FIRST.** Place the transfer, cover it with parchment, press about 12 seconds at medium-high pressure. Let it cool a few seconds, then peel the film.
- ☐ Cover it again and do a **SECOND** press of about 12 seconds — it makes the print softer and far more durable.
- ☐ Gently stretch the test print. If any edge lifts, press again a little hotter before you touch a real shirt.

30-90 min

- ☐ Run it as a line: one person measures and places, the adult presses, one person peels and does the second press, one person checks and folds.
- ☐ Measure placement on **EVERY** shirt. Eyeballing is how one shirt ends up with a crooked logo forever. Rotate every job except the press.

90-100 min

- ☐ Turn the press off first. Count finished shirts against the TS-3 order sheet, label them by size, and write down any spoiled transfer so a replacement can be ordered.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Test shirt pressed and passed the stretch check
- ☐ Every shirt pressed with each transfer in its measured spot
- ☐ Count matches the order sheet, spoiled transfers recorded
- ☐ Press turned off and cooling before cleanup

WHAT YOU NEED

- The club heat press (ask the coach where it is kept)
- Blank shirts and DTF transfers (TS-4)
- The placement notes from TS-2
- A spare shirt for testing
- Parchment or a Teflon sheet
- Heat-resistant gloves
- Ruler or tape measure
- An adult to run or supervise the press

BEFORE YOU START

TS-4

THIS UNLOCKS

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IF YOU FINISH EARLY

- Write the settings that worked — temperature, time, pressure, peel — on a card that lives with the press. Next season's team starts where you finished instead of guessing.

HAND OFF TO

Shirts go to the coach for handing out. The settings card stays with the press.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Fundraising is not asking strangers. It is asking the right people, and the right people are almost always connected to somebody already on the team.

YOU WILL FINISH WITH

25 or more real prospects with a reason and a personal connection noted for each, ranked A/B/C.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Get the fundraising goal from the coach and write it on the board. Everything today is measured against that number.

15-40 min

- ☐ Everyone lists out loud: where their parents work, businesses their family uses regularly, local engineering / tech / manufacturing / medical companies, and businesses within a mile of the school.

40-70 min

- ☐ Build the sheet with columns: business name, what they do, why they would care about robotics, who on the team has a connection, contact info if known.

70-90 min

- ☐ Rank each one A (a real personal connection), B (good fit but no connection), or C (long shot).

90-100 min

- ☐ Hand the list to the coach. **DO NOT CONTACT ANYONE YET** — the coach approves all outreach first.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Laptop or shared sheet
- Whiteboard
- The team's fundraising goal (ask the coach)

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

MF-2

IF YOU FINISH EARLY

- Check whether any of them publish a community-giving page, grant program, or employee matching program.

DONE WHEN – ALL 4 MUST BE TRUE

- ☐ 25+ prospects listed
- ☐ Connection and reason columns filled for each
- ☐ Ranked A/B/C
- ☐ Handed to the coach with nobody contacted

HAND OFF TO

The coach decides who gets contacted and how. MF-2 builds the thing you send them.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

A sponsor needs to know in 30 seconds who you are, what their money does, and what they get back. If it takes longer than one page, it does not get read.

YOU WILL FINISH WITH

A one-page sponsorship document with specific, deliverable tiers, reviewed by the coach.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-20 min**

- ☐ Gather the facts: team number 22257, school, number of students, what FIRST Tech Challenge is, what season it is, and what a season actually costs. Get REAL numbers from the coach — made-up numbers are worse than no numbers.

20-50 min

- ☐ Draft the page in four blocks: who we are (3 sentences), what we do, what we need, what you get.

50-75 min

- ☐ Build the tiers. Be concrete at each dollar level: logo on the shirt, logo on the robot, name on the website, a team visit to your office, a framed thank-you.
- ☐ Do not promise anything the team cannot actually deliver. A broken promise to a sponsor ends the relationship.

75-90 min

- ☐ Add a photo and the team contact — team email or the coach. NEVER a student's personal contact.

90-100 min

- ☐ Coach reviews before this leaves the room.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ Fits on one page
- ☐ Tiers are specific and genuinely deliverable
- ☐ Real budget numbers used
- ☐ No student personal contact info
- ☐ Coach reviewed

WHAT YOU NEED

- Identity brief and logo (EP-1)
- Real season budget numbers (ask the coach)
- A good team photo
- Design software or a word processor

BEFORE YOU START

EP-1, MF-1

THIS UNLOCKS

MF-3, MF-7

IF YOU FINISH EARLY

- Make a matching half-page version to hand out at events and demos.

HAND OFF TO

MF-3 writes the emails that carry this. MF-7 reuses the language for grant applications.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

The ask is the hardest part. Writing it together as a team makes it better and takes most of the fear out of it.

YOU WILL FINISH WITH

Three coach-approved templates — cold email, warm introduction, and thank-you — plus a tracking sheet.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Read the MF-2 one-pager so the emails match what it promises.

15-55 min

- ☐ Draft all three. Keep each UNDER 200 WORDS. Each one says: who we are, what we are asking for, what they get, and one clear next step.

55-75 min

- ☐ Read each one out loud. Anything that sounds stiff or fake, rewrite in normal words. Delete every sentence that is not doing work.

75-90 min

- ☐ Coach edits and approves. NOTHING gets sent by a student without coach approval — that is a hard team rule.

90-100 min

- ☐ Save the approved templates and build a tracking sheet: who, date contacted, response, follow-up date.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Sponsorship one-pager (MF-2)
- Laptop
- Coach available to approve

BEFORE YOU START

MF-2

THIS UNLOCKS

IF YOU FINISH EARLY

- Write the 30-second spoken version you would use if you walked into the business in person.

DONE WHEN – ALL 4 MUST BE TRUE

- ☐ 3 templates, each under 200 words
- ☐ Coach approved
- ☐ Tracking sheet created
- ☐ Nothing sent yet

HAND OFF TO

The coach decides who sends what and when. Keep the tracking sheet current.

Who worked on this: _____ Date: _____

Social Media Plan & First Posts

WHY THIS MATTERS

Sponsors, judges, and future team members all look you up. A page with three good posts beats a page with nothing, and it beats an abandoned page with one post from October.

YOU WILL FINISH WITH

Written posting rules, three coach-approved posts ready to go, and a named owner.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Confirm with the coach which accounts the team may use and write down the rules: no student full names or faces without permission on file, no personal contact info, coach approves every post before it goes up.

20-40 min

- ☐ Plan the rhythm: what gets posted (build progress, competition results, sponsor thank-yous, member spotlights) and how often. Pick something you can actually sustain.

40-80 min

- ☐ Draft 3 real posts — caption plus the specific photo. Take the photos today if you do not already have them.

80-95 min

- ☐ Coach reviews every caption AND every photo, including who is visible in each one.

95-100 min

- ☐ Schedule them or hand them off. Write down who owns posting each week.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Identity brief (EP-1)
- Camera or phone
- Coach available to approve
- Access to whichever accounts the team uses

BEFORE YOU START

EP-1

THIS UNLOCKS

IF YOU FINISH EARLY

- Write the sponsor thank-you post template now, so it is ready the moment a sponsor says yes.

DONE WHEN – ALL 4 MUST BE TRUE

- ☐ Photo and privacy rules written down
- ☐ 3 posts drafted with photos attached
- ☐ Coach approved every post and photo
- ☐ Weekly owner named

HAND OFF TO

Give the rules to everyone who takes photos at meetings and events.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

The team needs a page on the Herobotix site, and whoever has website access can only publish what you hand them. Someone has to actually write it.

YOU WILL FINISH WITH

A finished content package – copy plus images plus instructions – in one folder, ready to hand off.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Look at the existing Herobotix website and the other teams' pages. Match their format so it drops in cleanly.

15-50 min

- ☐ Write the copy: team number and name, a two-paragraph 'who we are', what this season's game is, meeting times, how to join, how to sponsor.

50-75 min

- ☐ Gather the images. CHECK WITH THE COACH that every student pictured has photo permission on file.

75-90 min

- ☐ Proofread twice. Verify every fact: team number, dates, and the spelling of every name.

90-100 min

- ☐ Put the text and images in ONE clearly named shared folder with a short note saying what goes where.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ Copy written and proofread twice
- ☐ Images collected with photo permissions verified
- ☐ Everything in one named folder with a placement note

WHAT YOU NEED

- Identity brief and logo (EP-1)
- Team photo and robot photo
- Laptop
- Photo permission list from the coach

BEFORE YOU START

EP-1

THIS UNLOCKS

IF YOU FINISH EARLY

- Write a shorter version suitable for a school newsletter or a season kickoff announcement.

HAND OFF TO

Hand the folder to the coach, who passes it to whoever administers the Herobotix site.

Who worked on this: _____ Date: _____

Plan a Demo / Outreach Event

WHY THIS MATTERS

Outreach is judged at competition, it recruits next year's team, and it is genuinely fun. Elementary schools lose their minds over a robot.

YOU WILL FINISH WITH

A complete 20-minute run-of-show with a materials list and assigned roles, ready to hand to the coach for scheduling.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Pick the audience and venue: elementary school, library, school open house, a sponsor's office. Check with the coach on what is actually schedulable.

20-50 min

- ☐ Design the 20 minutes: what you show, who says what, and what the audience gets to TOUCH or DRIVE. Make it hands-on — watching a robot is boring after 90 seconds.

50-70 min

- ☐ List everything you need: robot, charged batteries, charged spares, field elements, table, extension cord, give-aways, a team sign.

70-90 min

- ☐ Assign roles and write the run-of-show minute by minute, so nobody is standing around wondering what happens next.

90-100 min

- ☐ Write the request to the coach: proposed dates, what you need from adults, and how the robot gets there.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- A robot that drives (PR-3)
- Paper for planning
- Laptop
- Coach's input on what is realistic

BEFORE YOU START

PR-3

THIS UNLOCKS

IF YOU FINISH EARLY

- Write down 5 questions a 3rd grader would actually ask and the answers, so nobody freezes.

DONE WHEN – ALL 4 MUST BE TRUE

- ☐ Minute-by-minute 20-minute run-of-show written
- ☐ Complete materials list
- ☐ Roles assigned by name
- ☐ Written request handed to the coach

HAND OFF TO

The coach schedules it. Rehearse once before the real thing.

Who worked on this: _____ Date: _____

Grants & Team Funding Applications

No hardware

Challenge

2 people

100 min

WHY THIS MATTERS

FIRST and its sponsors offer grants that most teams never apply for, because nobody read the page. This is money that rewards paperwork.

YOU WILL FINISH WITH

A table of every grant the team qualifies for with deadlines, and one application draft started.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-25 min

- ☐ With the coach, go through the FIRST Dashboard and FIRST's grant and funding resources.
- ☐ Also check local sources: the school PTO, city or community foundations, and employer matching programs at parents' companies.

25-50 min

- ☐ Build the table: grant name, amount, deadline, who is allowed to apply (student or adult), and what the application requires.

50-60 min

- ☐ Sort by deadline. Flag anything due within the next 60 days in red.

60-95 min

- ☐ Take the nearest deadline and draft answers to its questions. Reuse the language from the MF-2 one-pager rather than starting from scratch.

95-100 min

- ☐ Hand the table and the draft to the coach — most applications must be submitted by an adult.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Grant table with amounts and deadlines
- ☐ Anything due within 60 days flagged
- ☐ One application draft started
- ☐ Handed to the coach

WHAT YOU NEED

- FIRST Dashboard access (with the coach)
- Sponsorship one-pager (MF-2)
- Laptop
- Coach working alongside you

BEFORE YOU START

MF-2

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Put a calendar reminder two weeks before every deadline on the list.

HAND OFF TO

The coach submits. Put the deadlines on the team calendar.

Who worked on this: _____ Date: _____